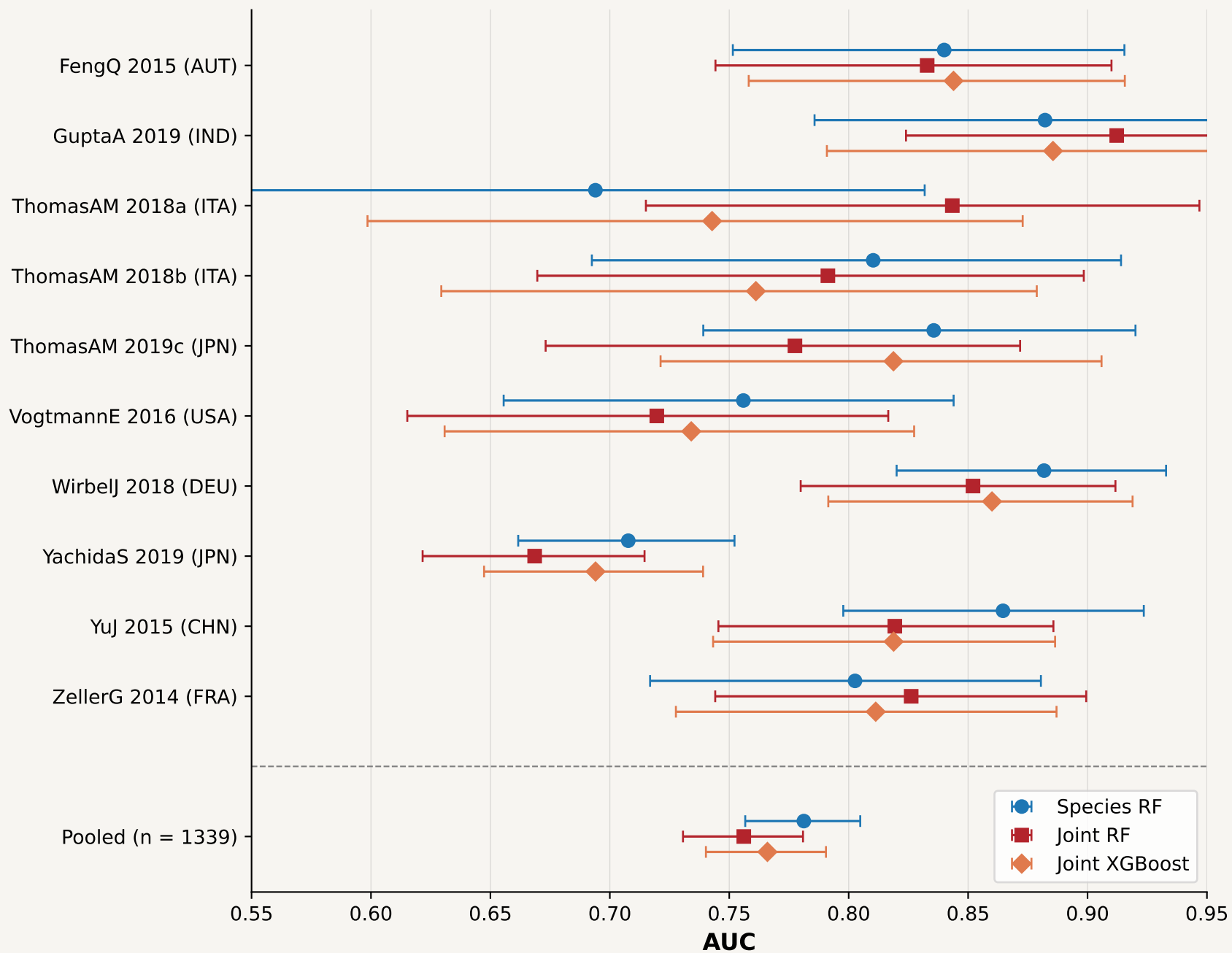
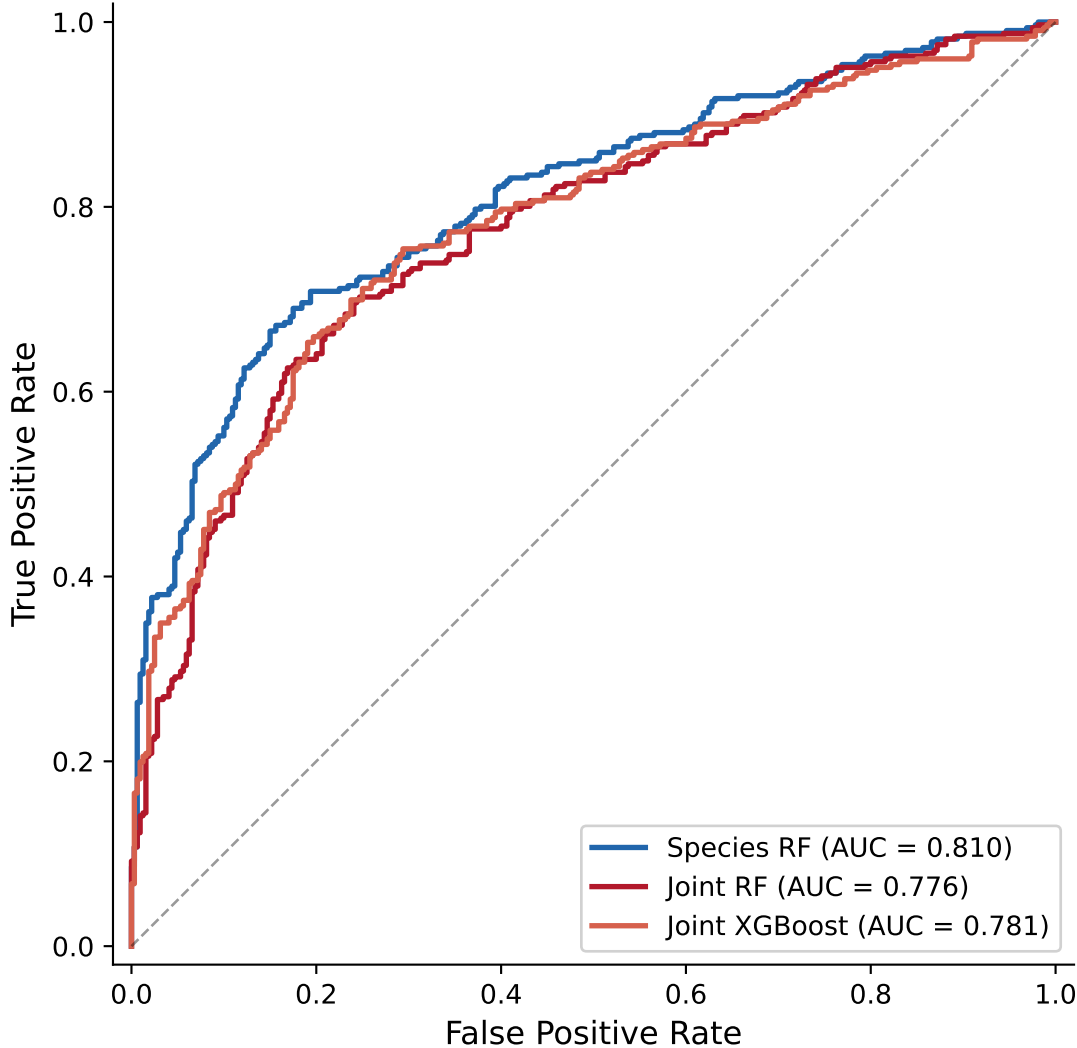


Figure 1. Per-cohort and pooled AUC with 95% bootstrap CI under LODO cross-validation

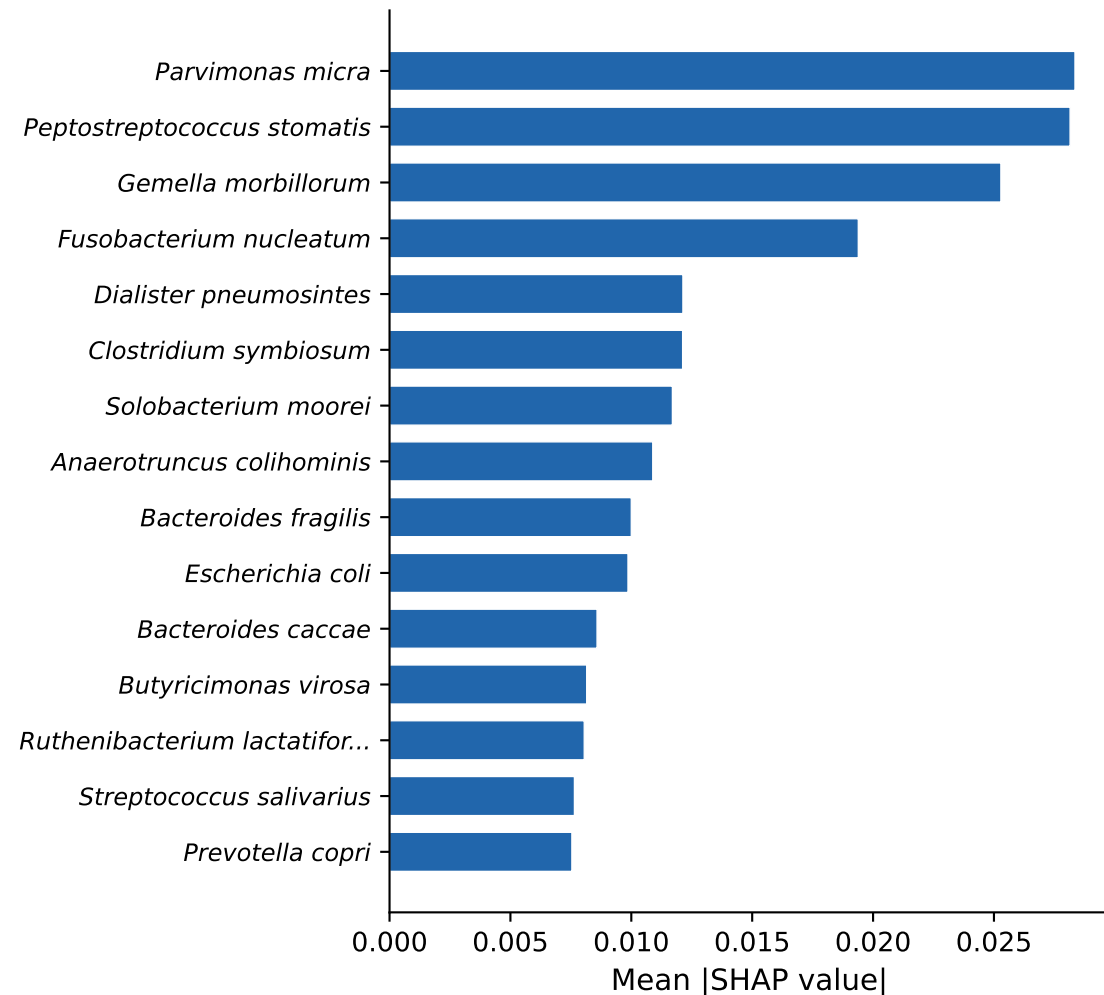


Pooled LODO ROC Curves (n = 646)

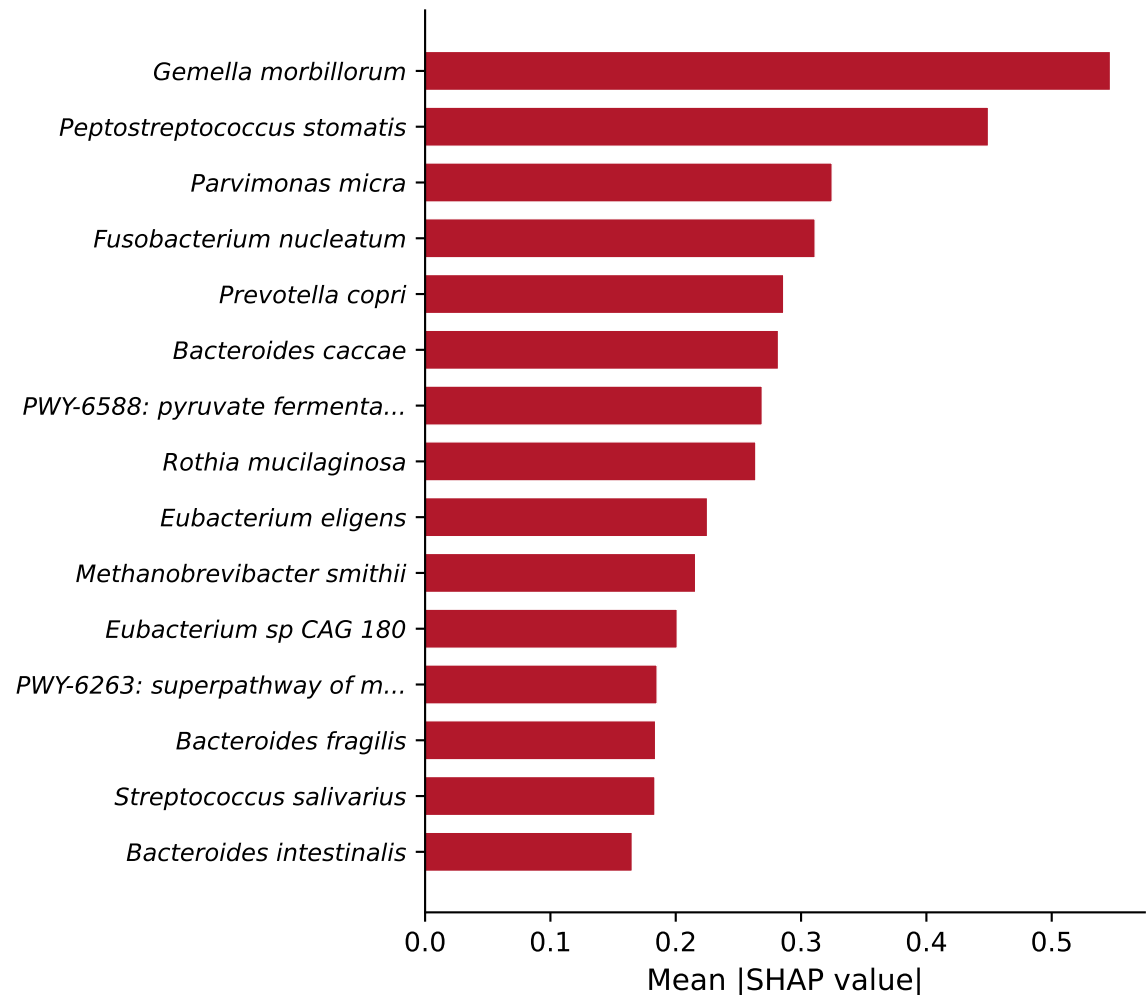


Top 15 Species by SHAP Feature Importance (CRC vs. Healthy)

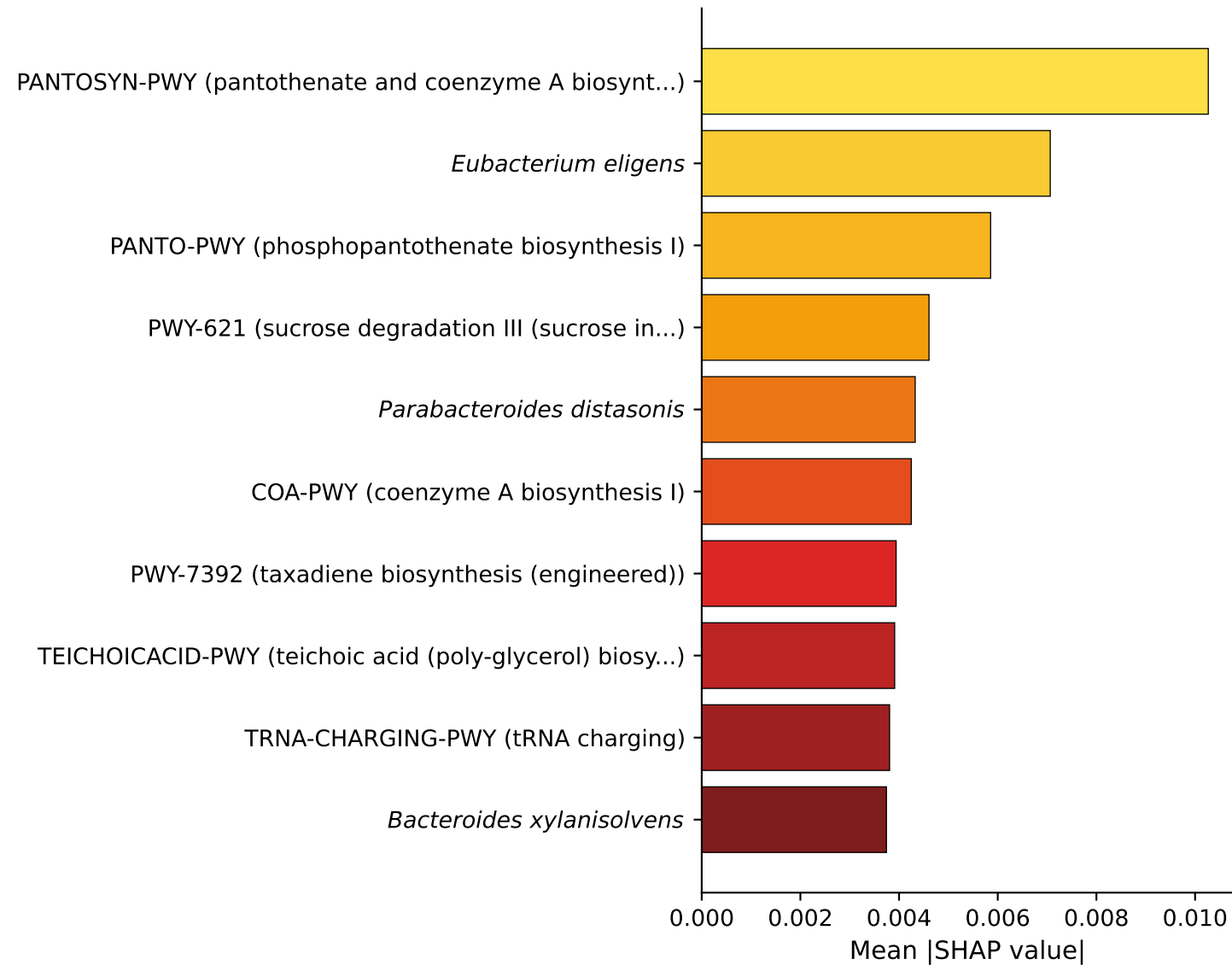
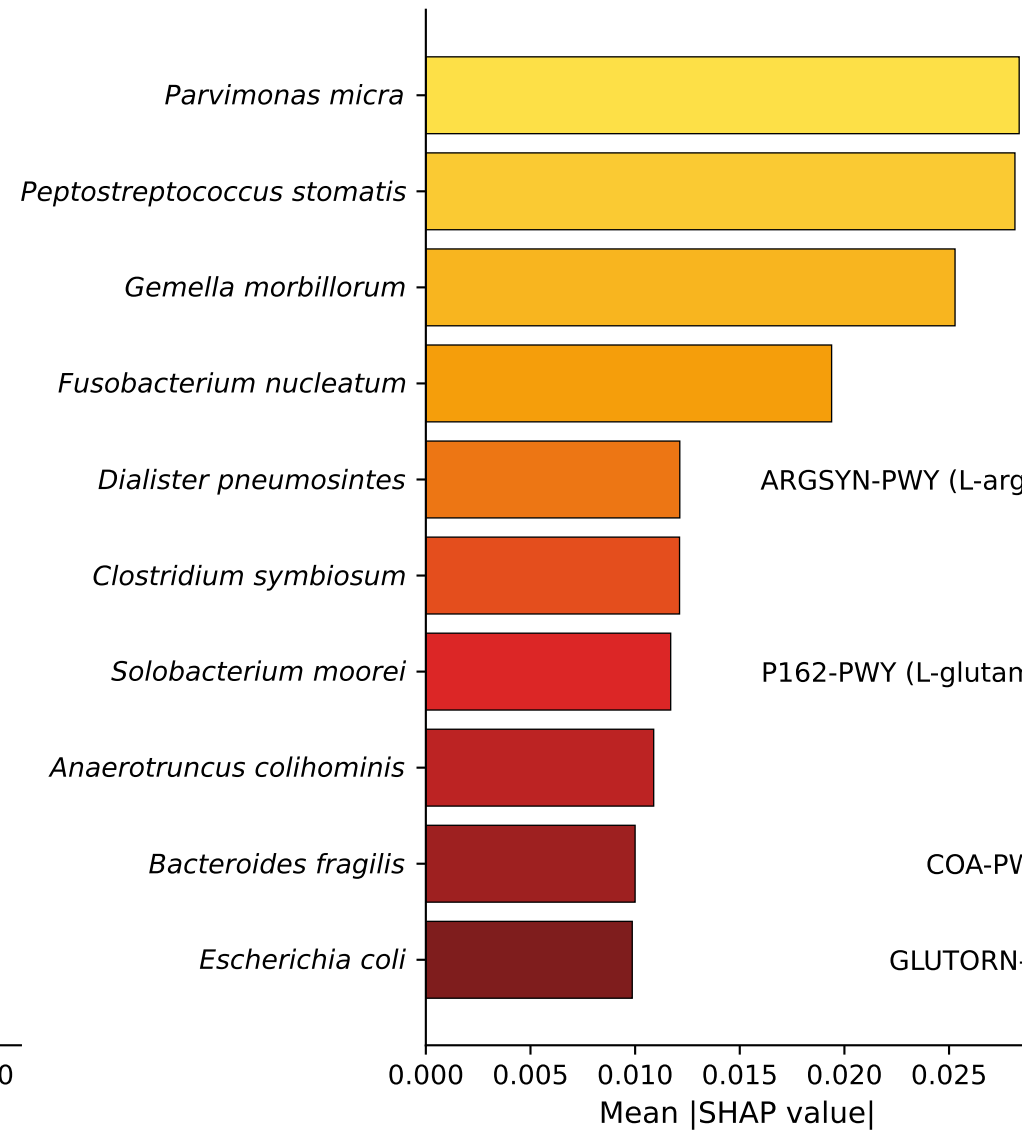
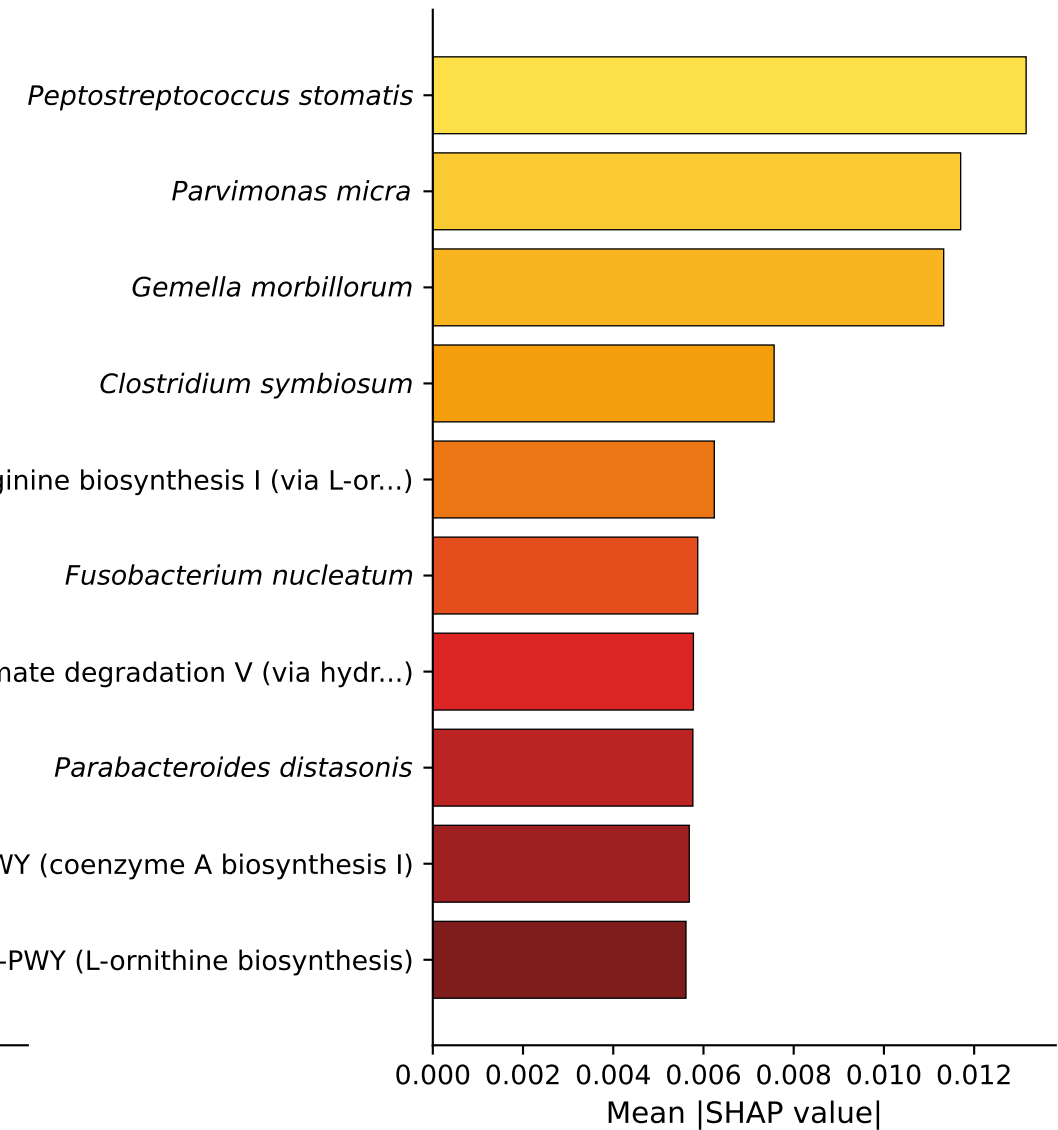
Random Forest



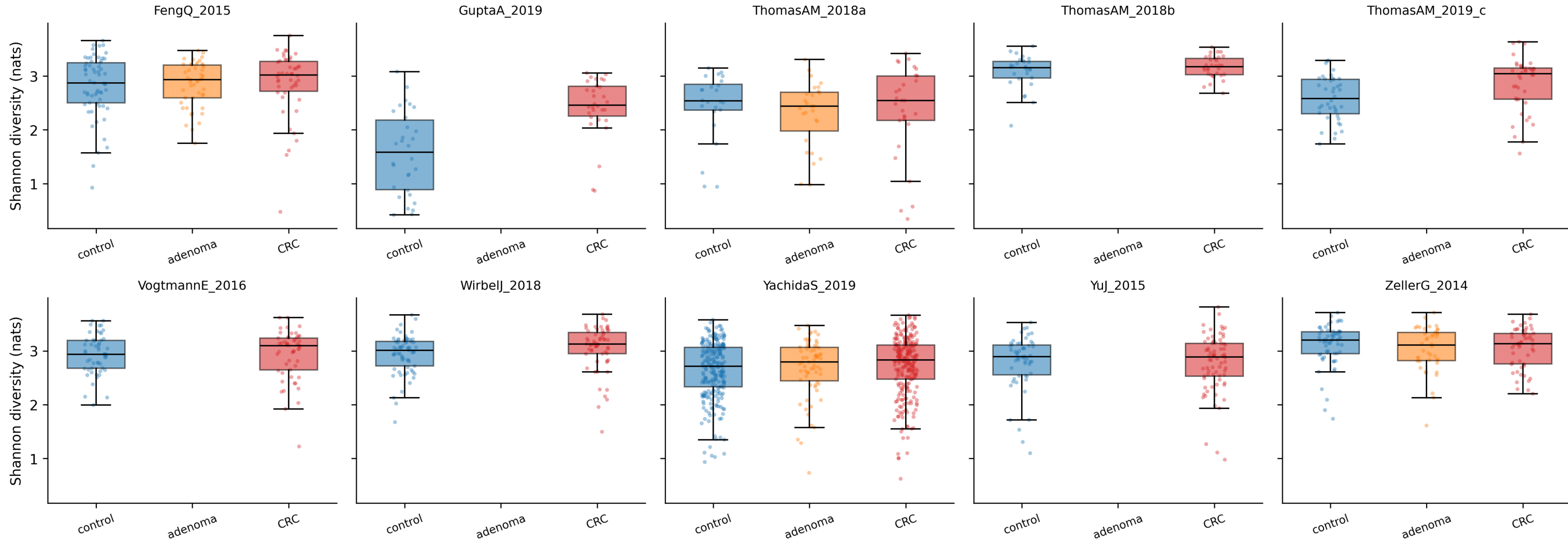
XGBoost



8/15 features shared in top 15 between RF and XGBoost

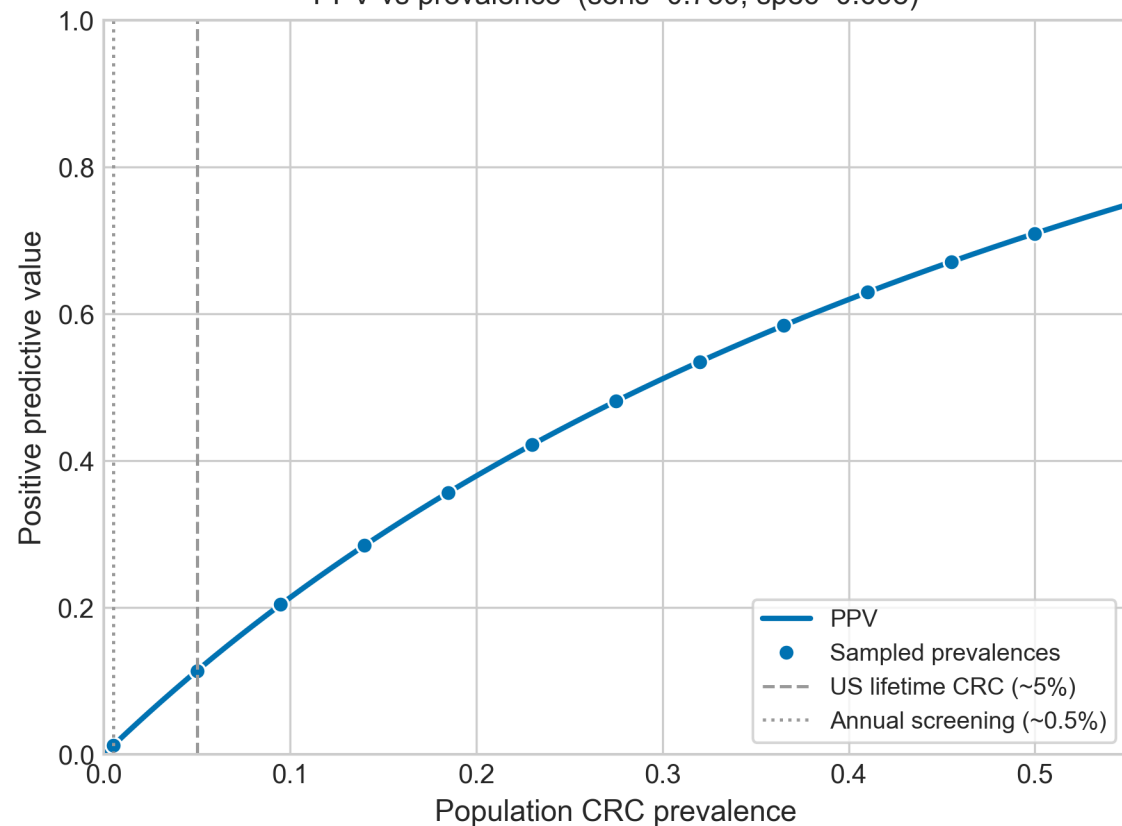
A**Healthy vs Adenoma
(commensal depletion phase)****B****CRC vs Healthy
(oral pathobiont colonization)****C****Adenoma vs CRC
(oral pathobiont colonization)**

Alpha diversity (Shannon) by study_condition, per cohort

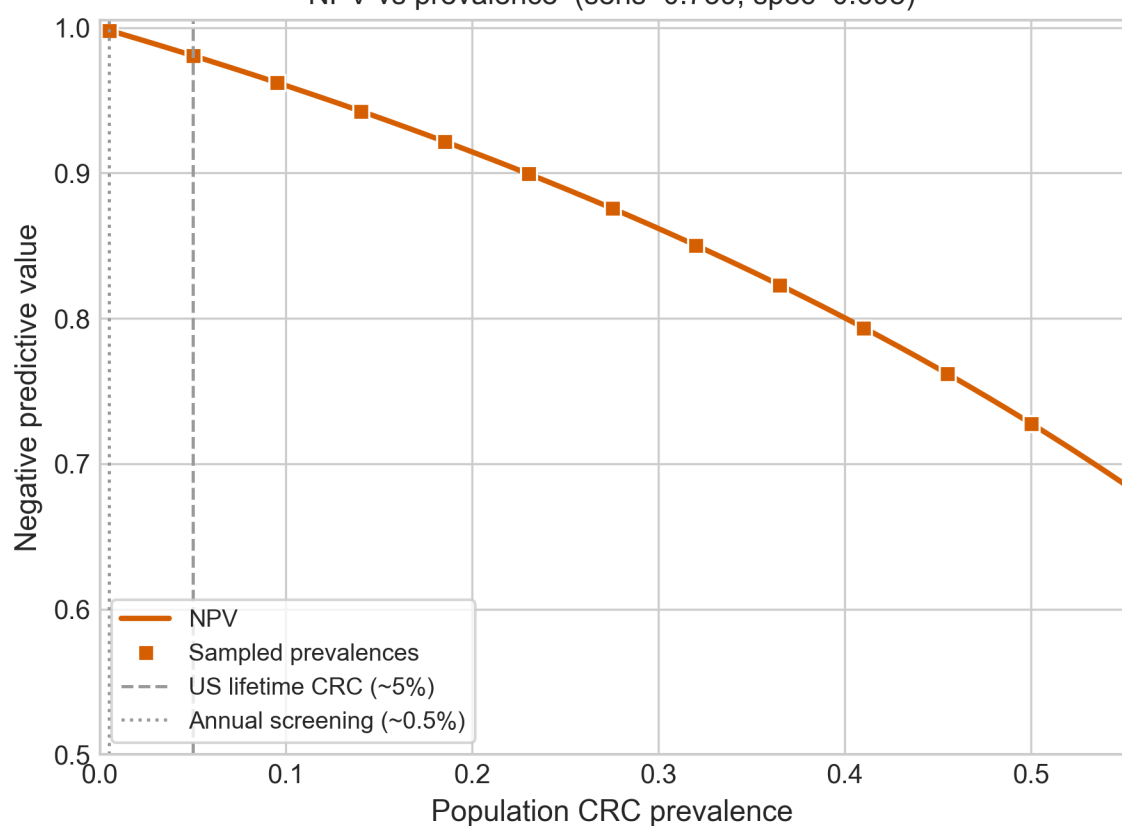


Species RF (Youden-J fixed): base-rate-adjusted PPV and NPV

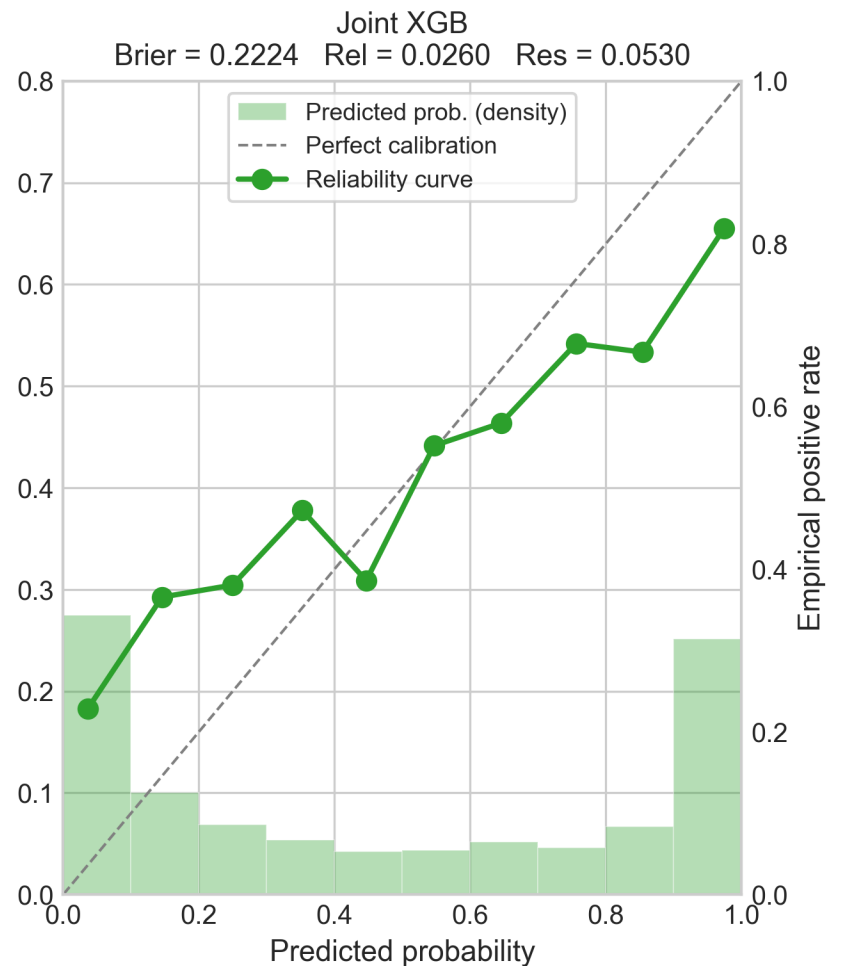
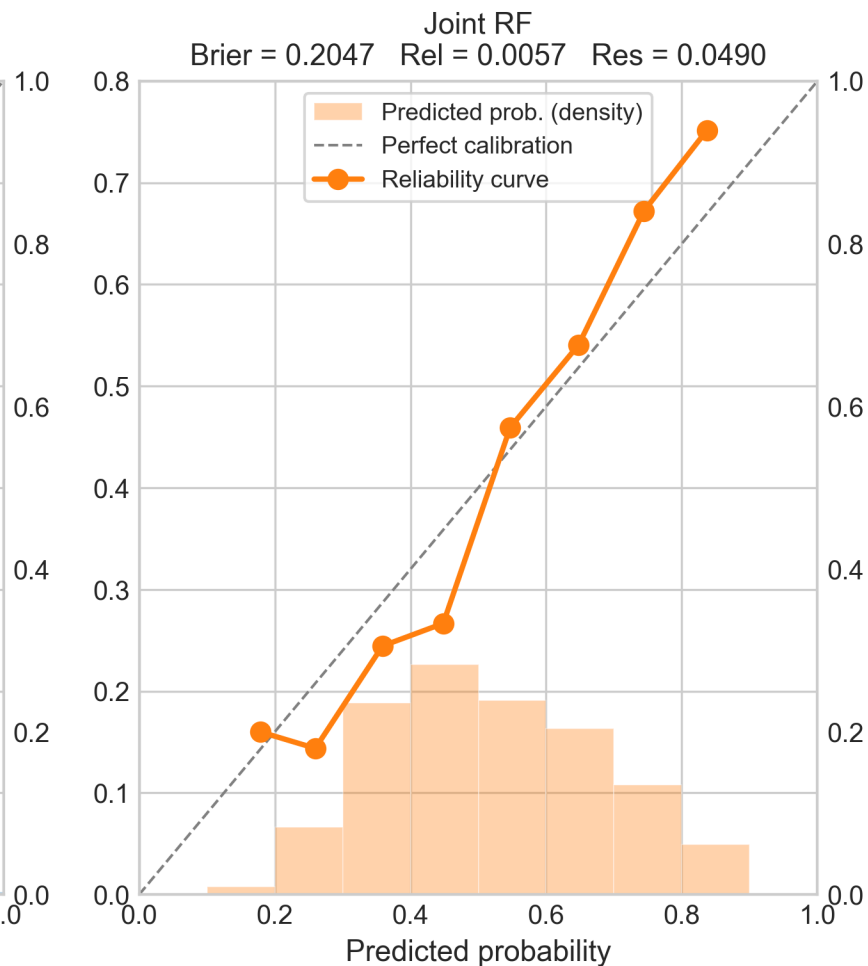
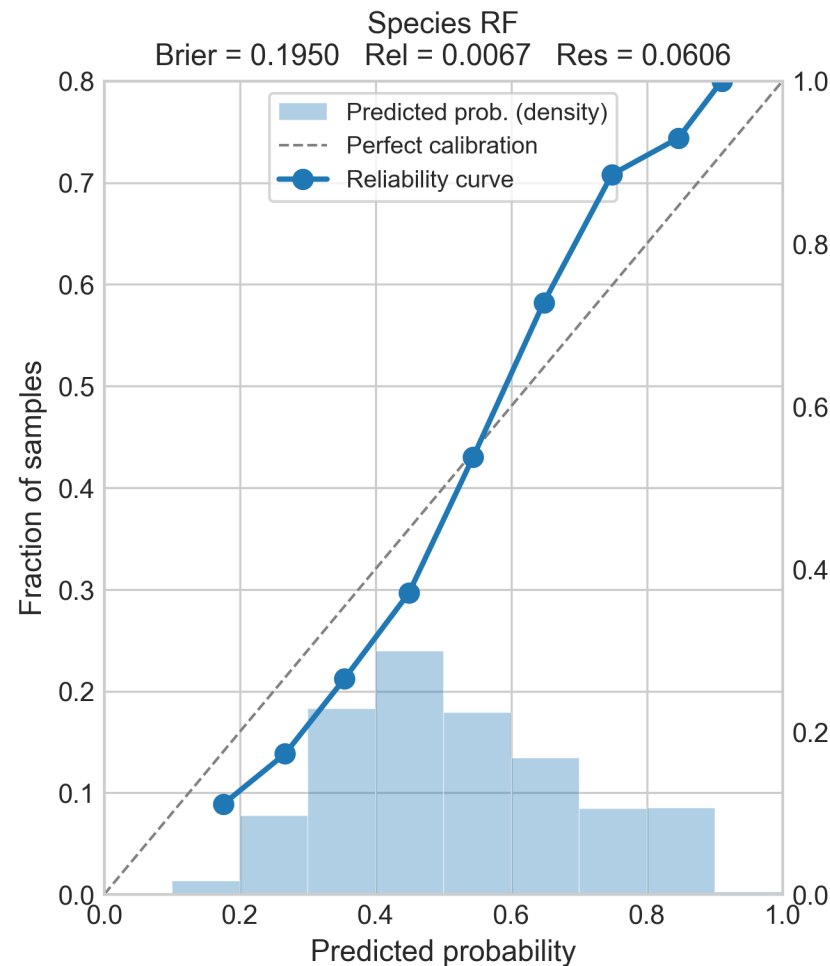
PPV vs prevalence (sens=0.739, spec=0.698)



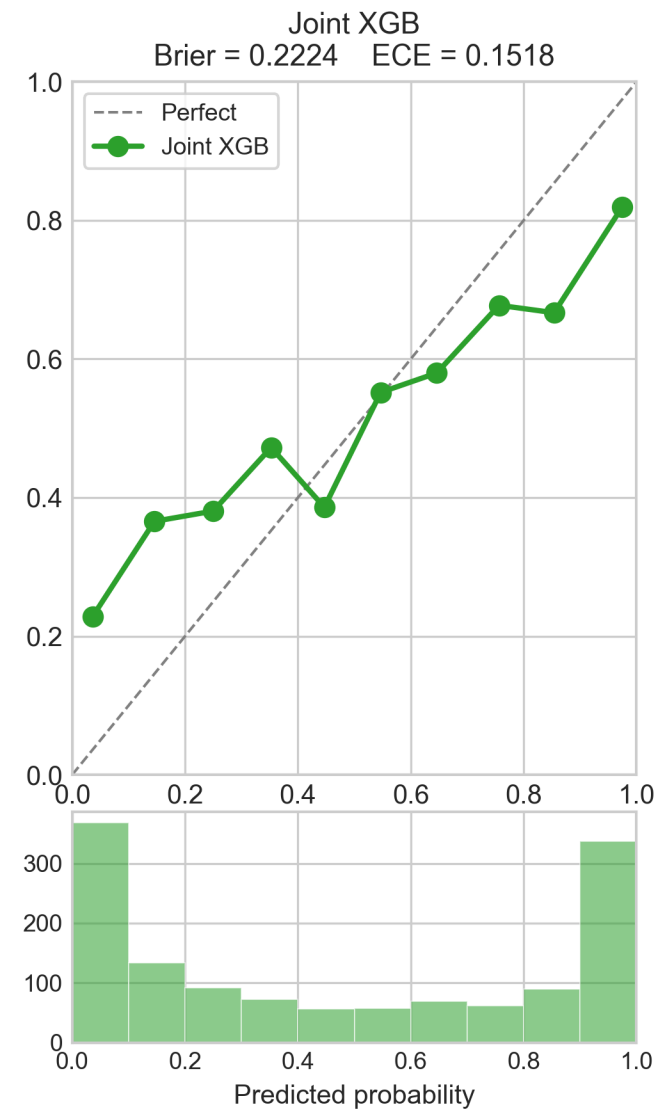
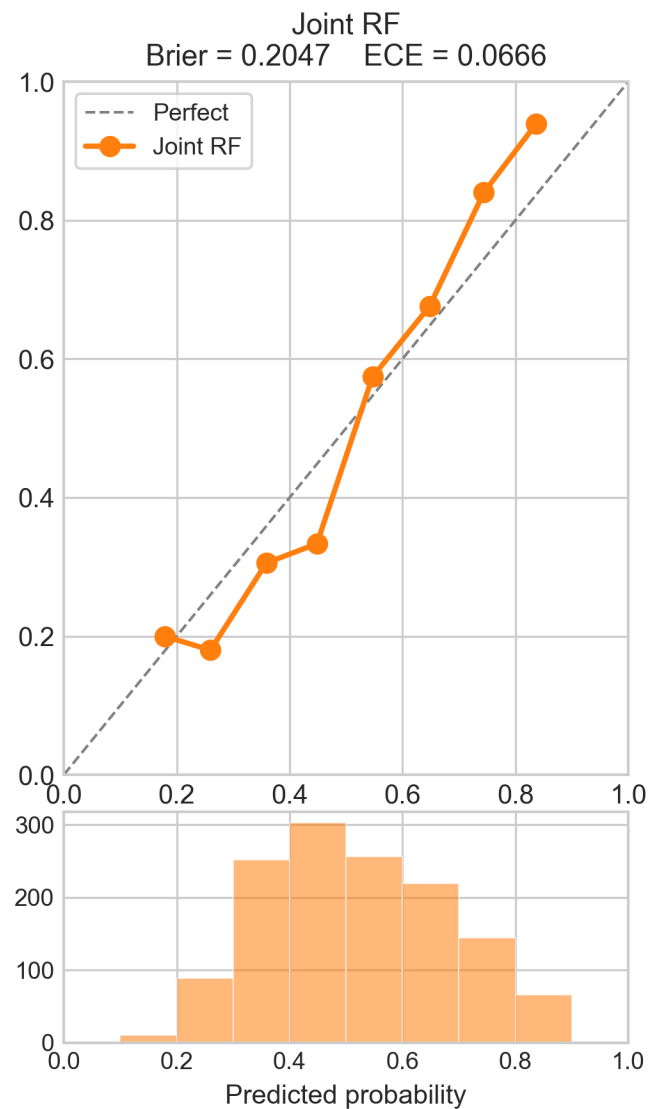
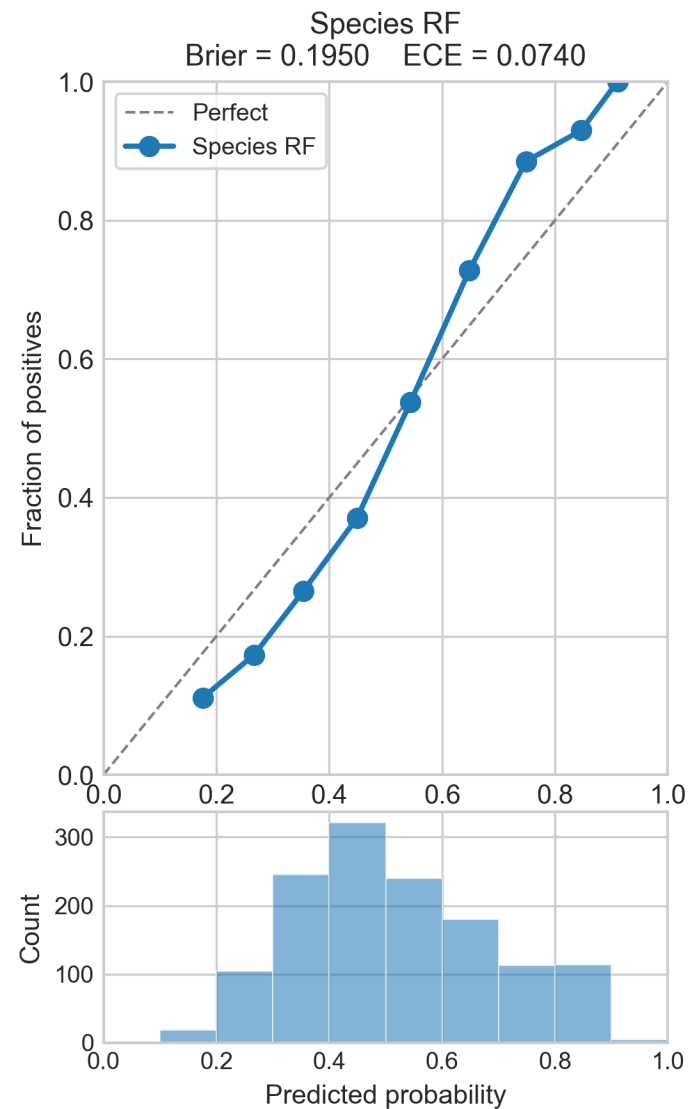
NPV vs prevalence (sens=0.739, spec=0.698)



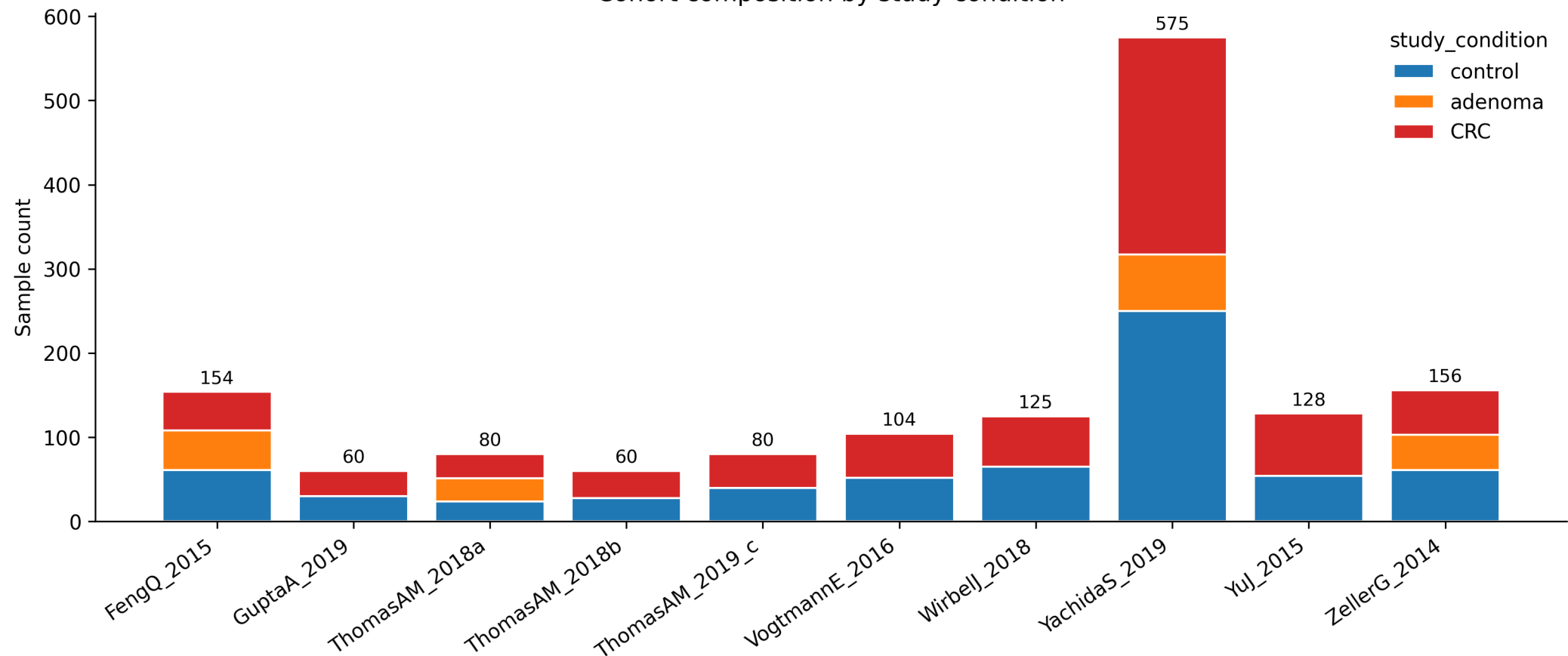
Calibration mechanism — joint XGB pushes probabilities to 0/1, inflating reliability term



Reliability diagrams (10-bin equal-width) — pooled LODO predictions

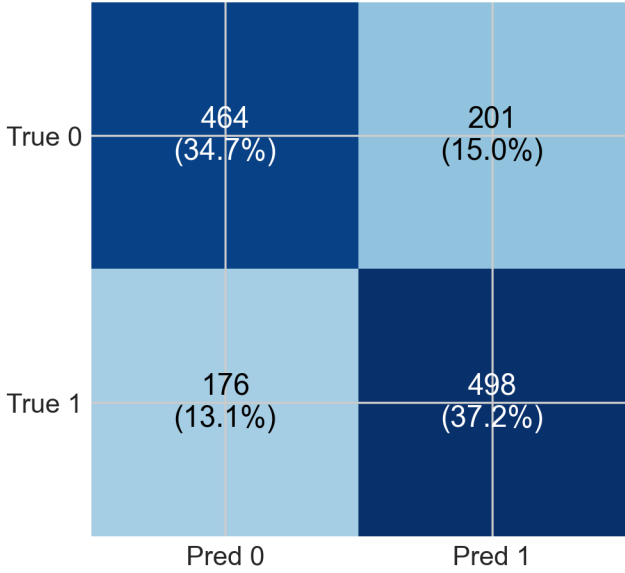


Cohort composition by study condition

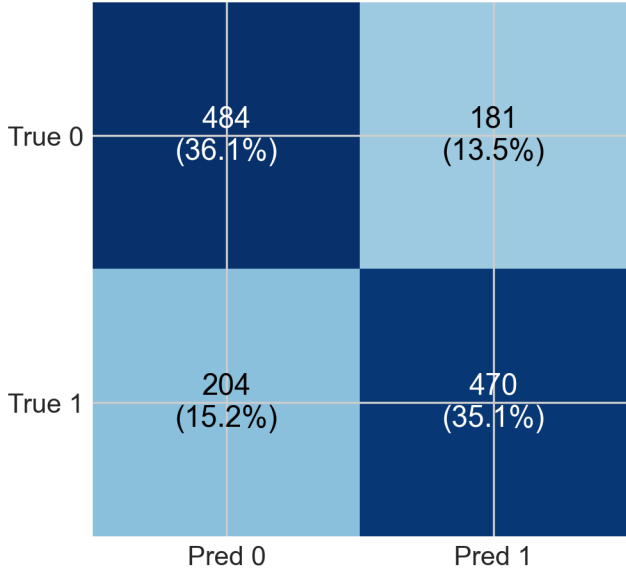


Confusion matrices — pooled LODO predictions

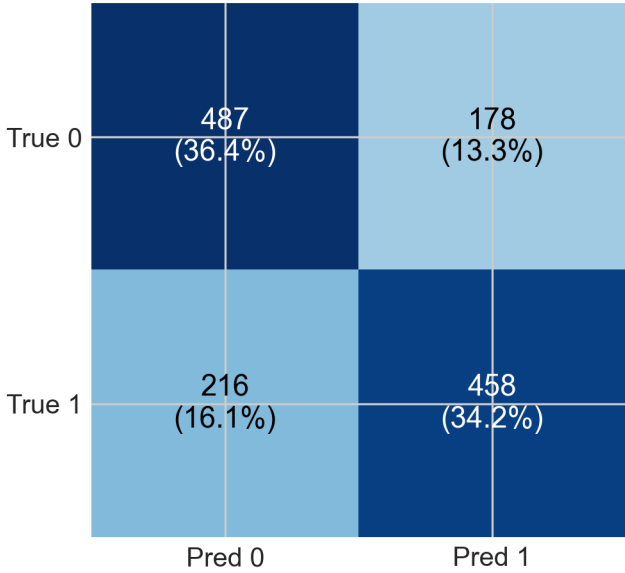
Species RF — Youden-J (t=0.485)
Sens=0.739 Spec=0.698 MCC=0.437



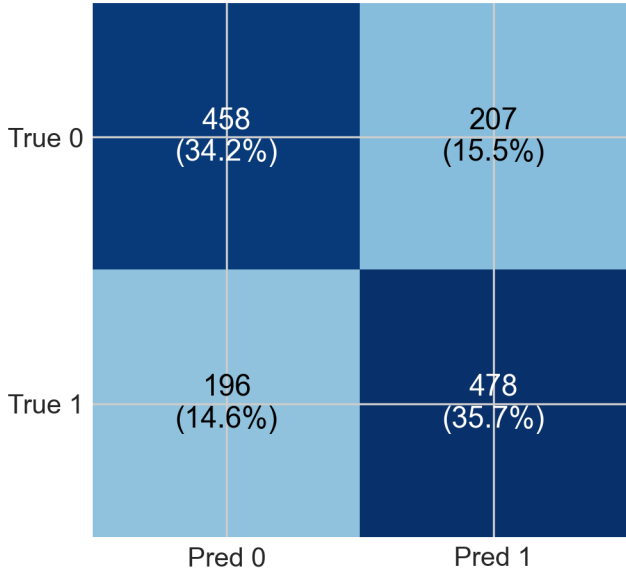
Species RF — Threshold = 0.5 (t=0.500)
Sens=0.697 Spec=0.728 MCC=0.425



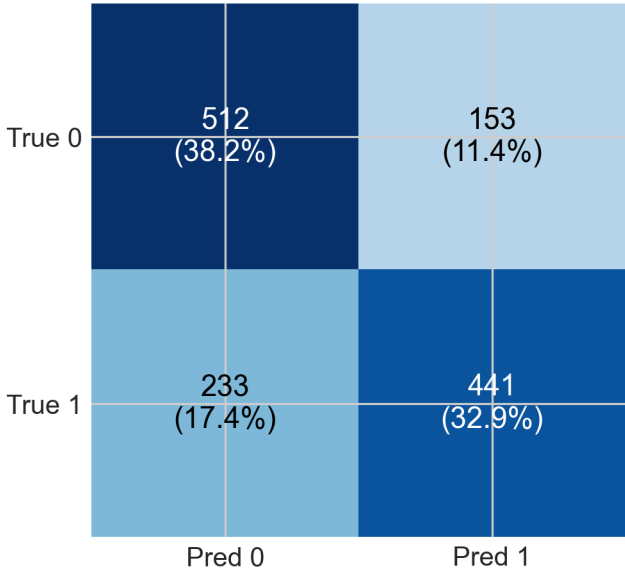
Joint RF — Youden-J (t=0.519)
Sens=0.680 Spec=0.732 MCC=0.412



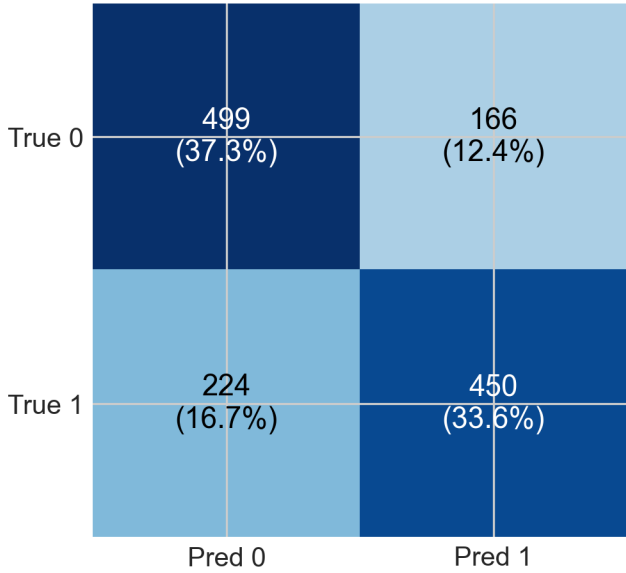
Joint RF — Threshold = 0.5 (t=0.500)
Sens=0.709 Spec=0.689 MCC=0.398



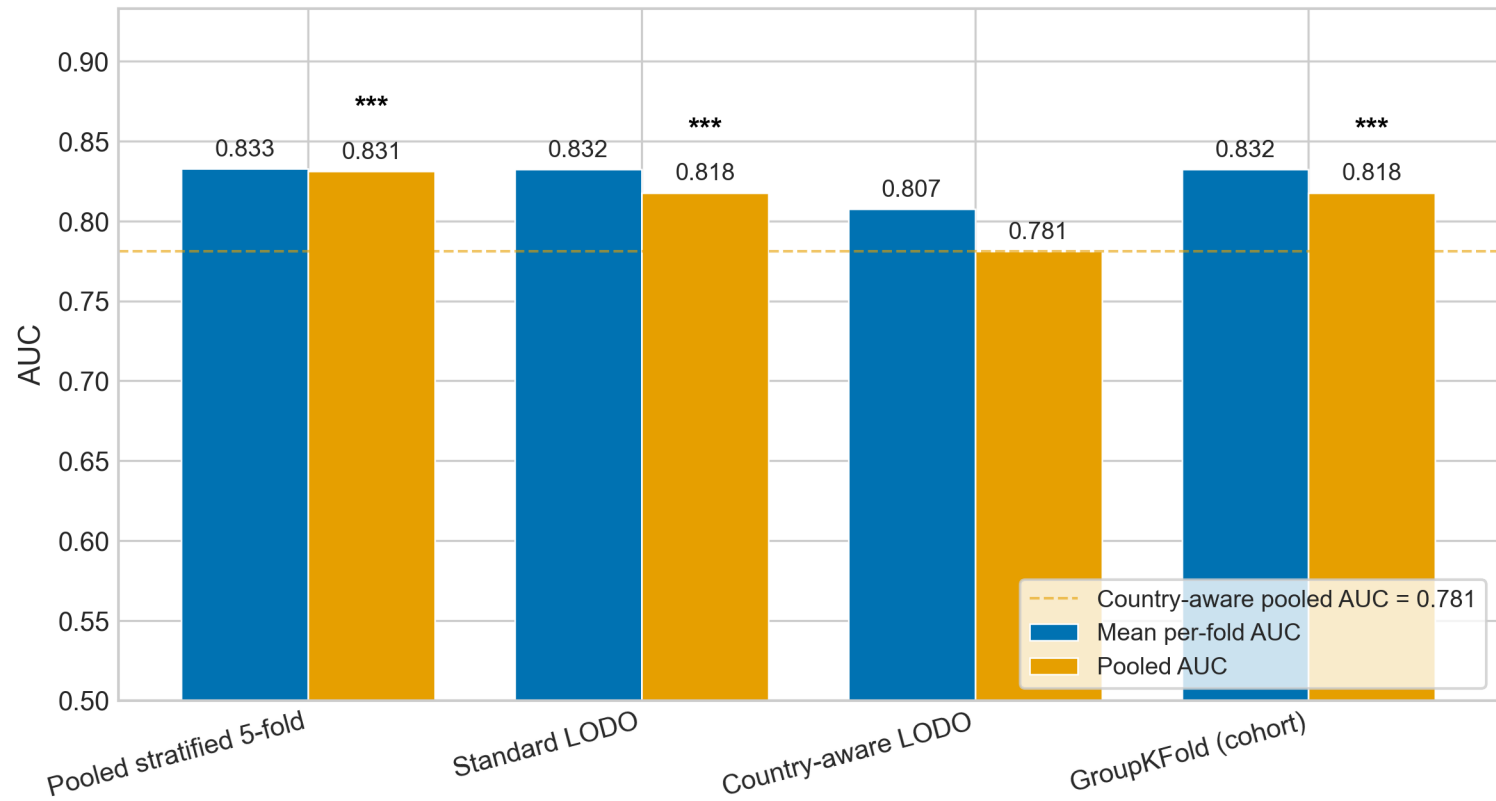
Joint XGB — Youden-J (t=0.533)
Sens=0.654 Spec=0.770 MCC=0.427



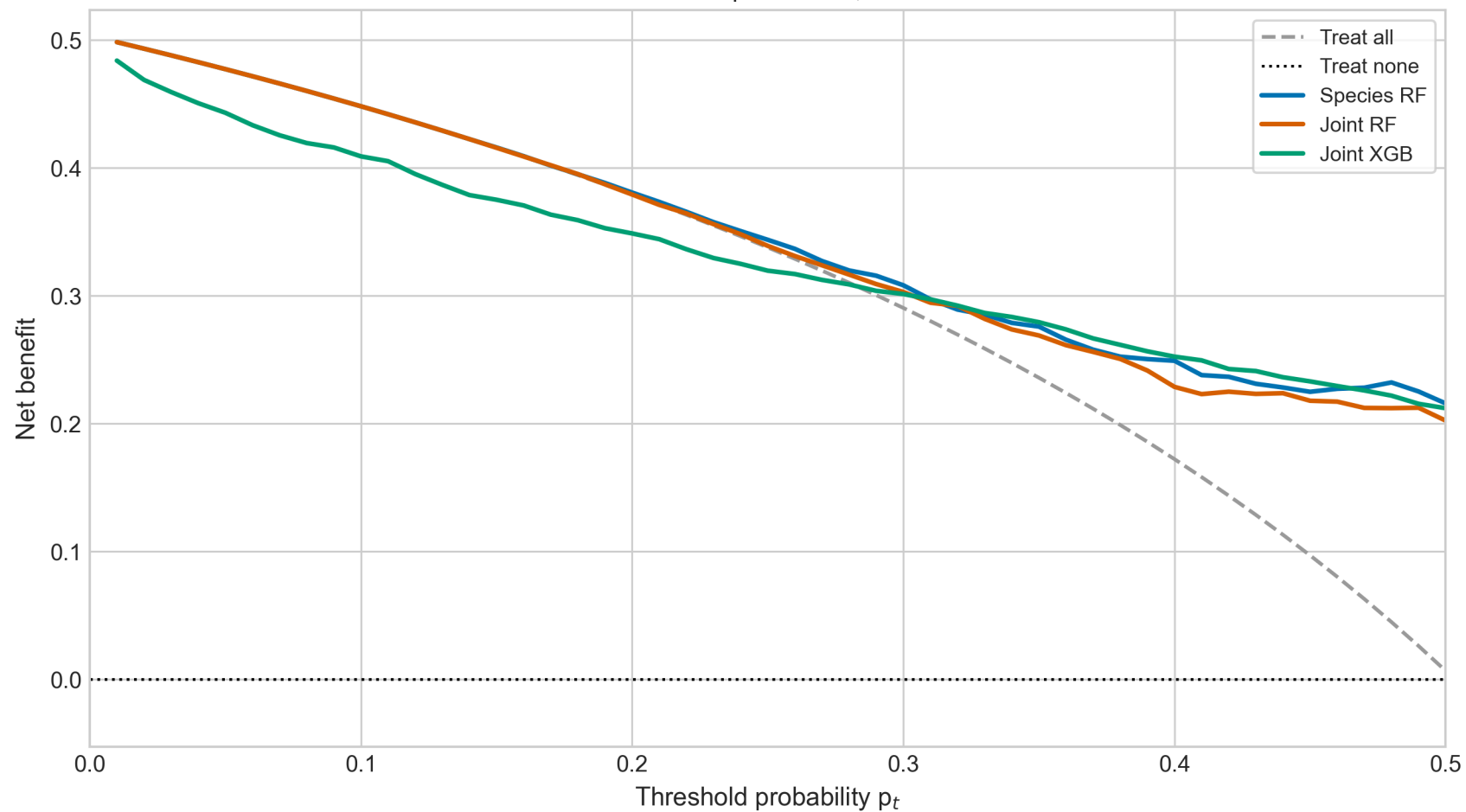
Joint XGB — Threshold = 0.5 (t=0.500)
Sens=0.668 Spec=0.750 MCC=0.419



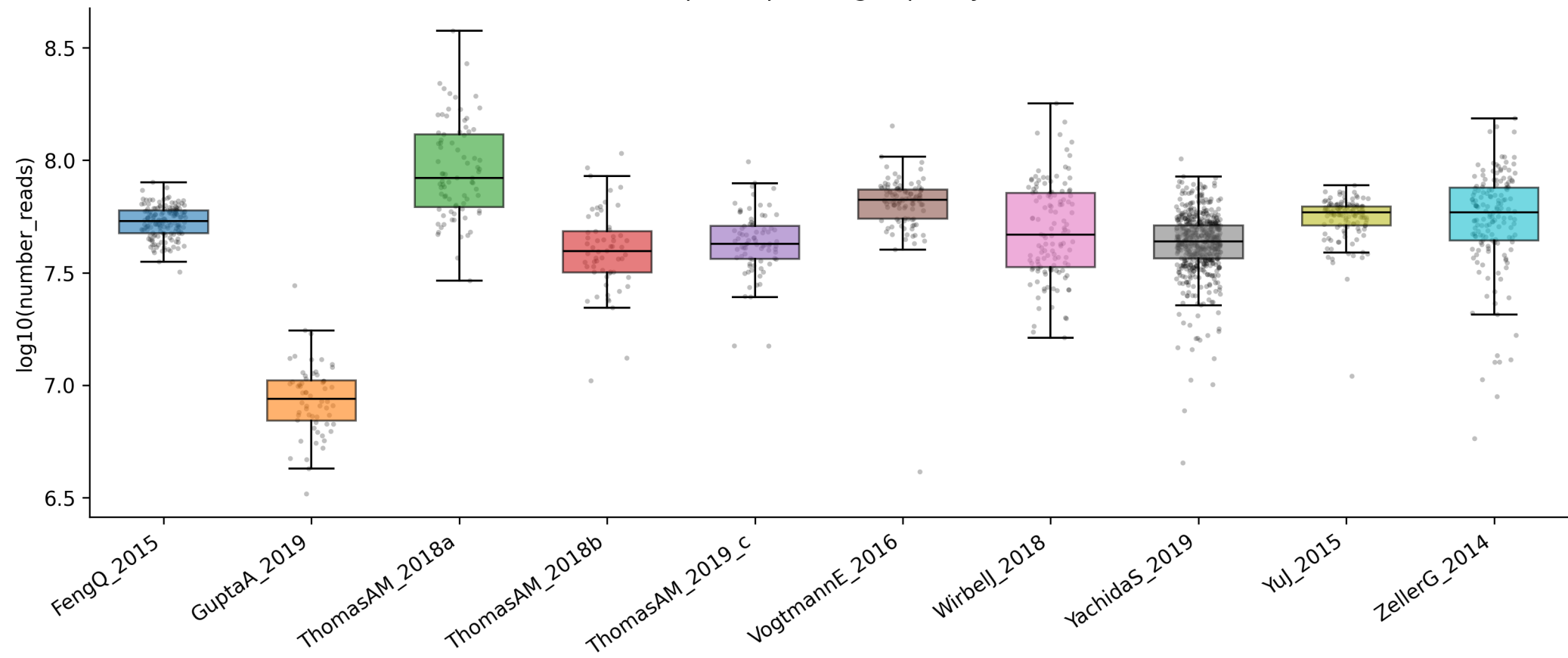
Cross-validation methodology comparison — species RF
(stars: DeLong p vs country-aware LODO on pooled predictions)



Decision-curve analysis (Vickers & Elkin)
Pooled LODO predictions, n = 1339

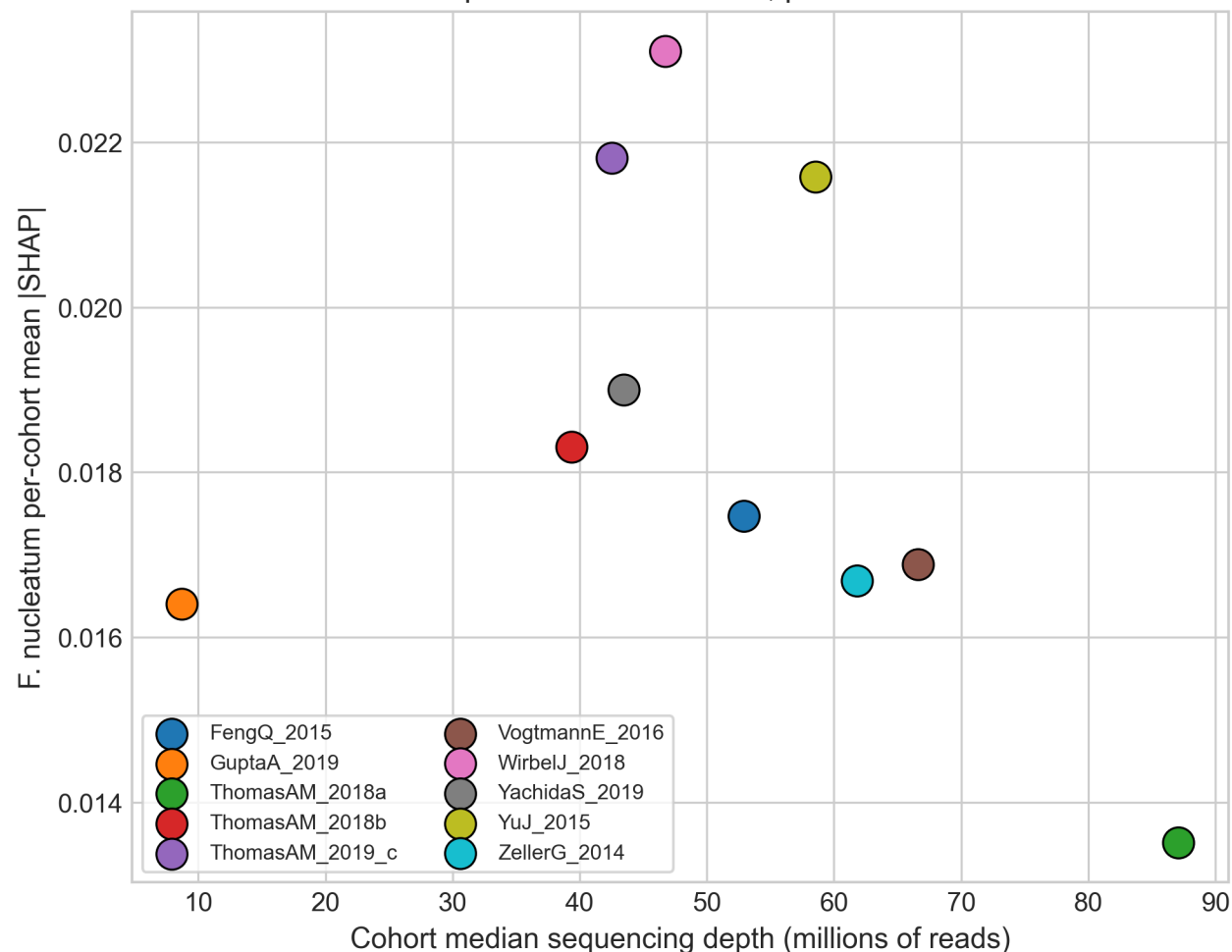


Per-sample sequencing depth by cohort

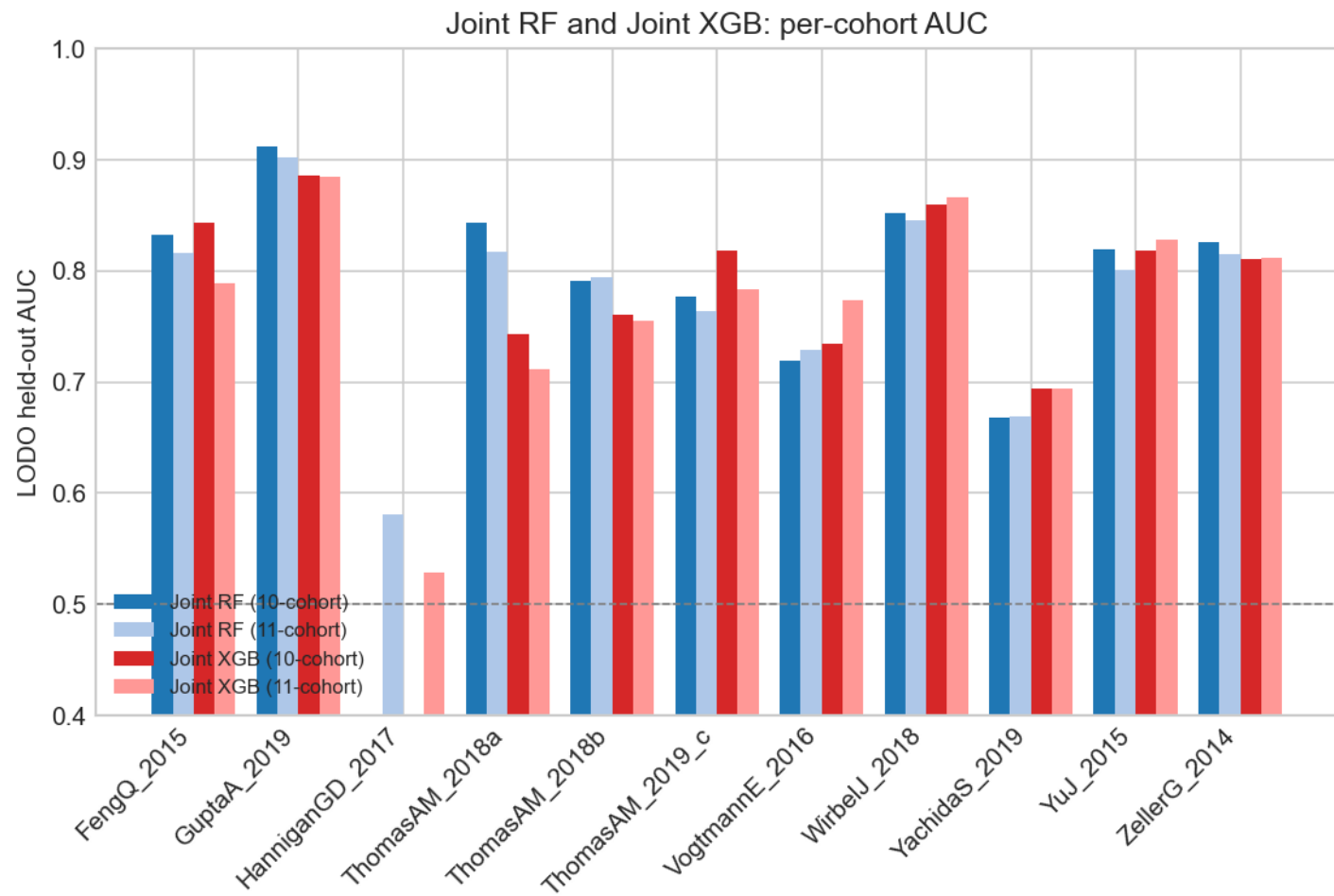
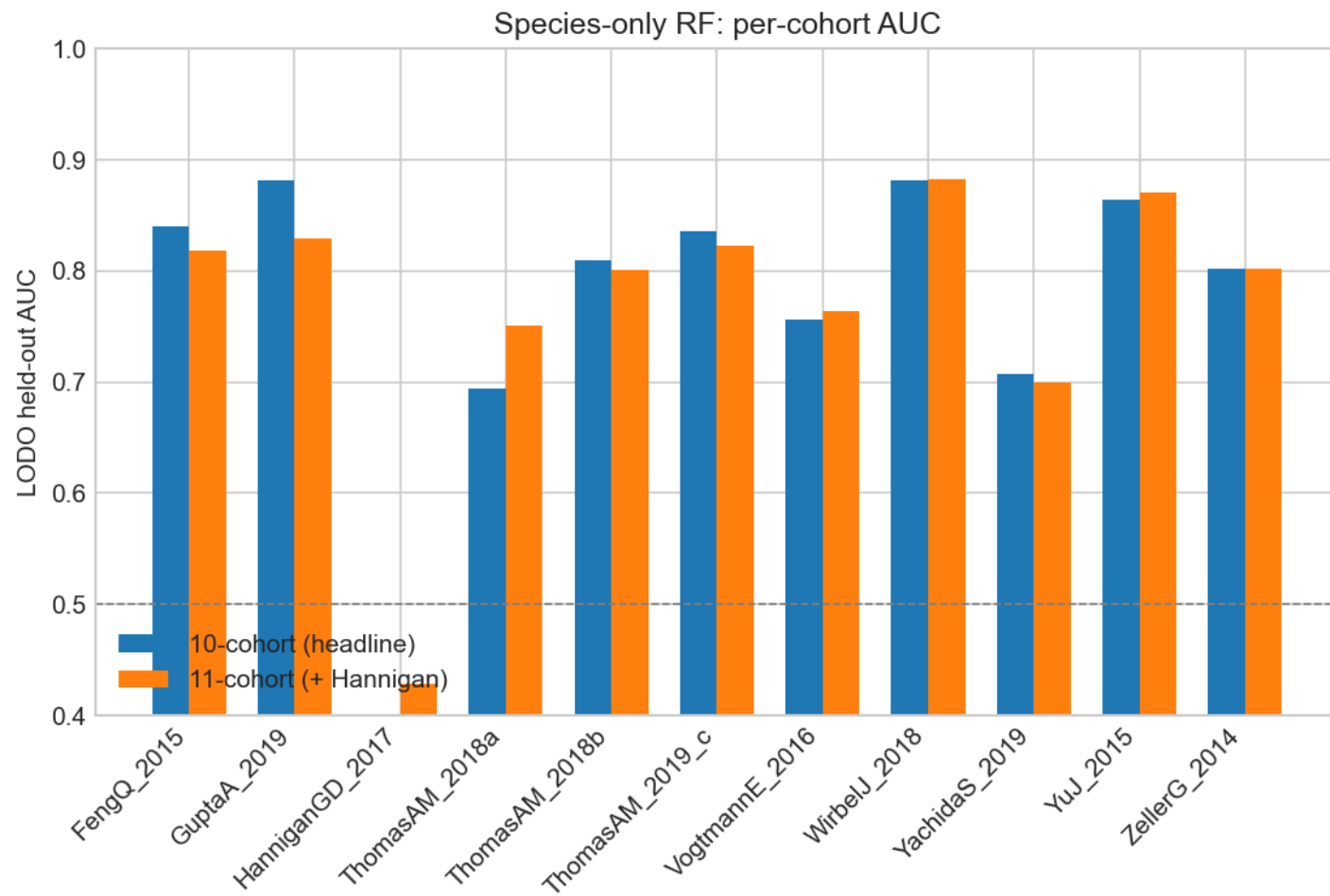


F. nucleatum SHAP vs cohort sequencing depth

Spearman rho = -0.194, p = 0.591

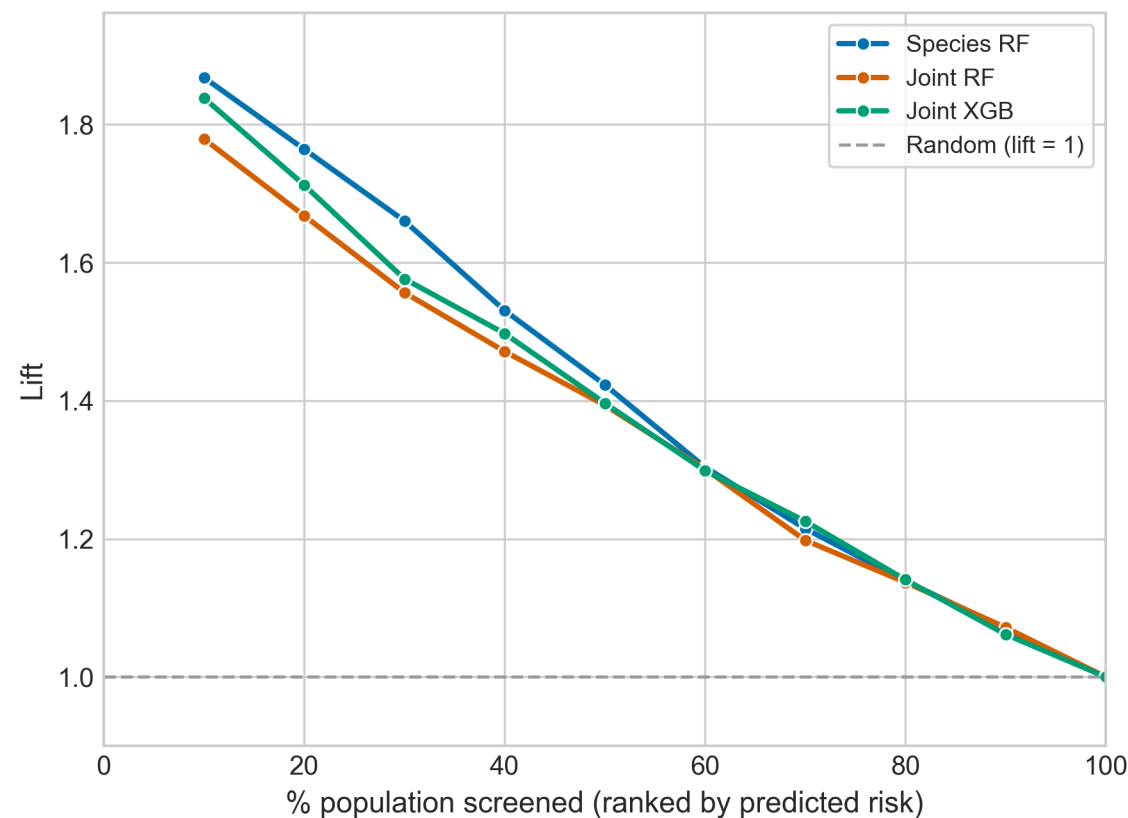


Cohort-exclusion sensitivity: 10-cohort headline vs 11-cohort (HanniganGD_2017 included)

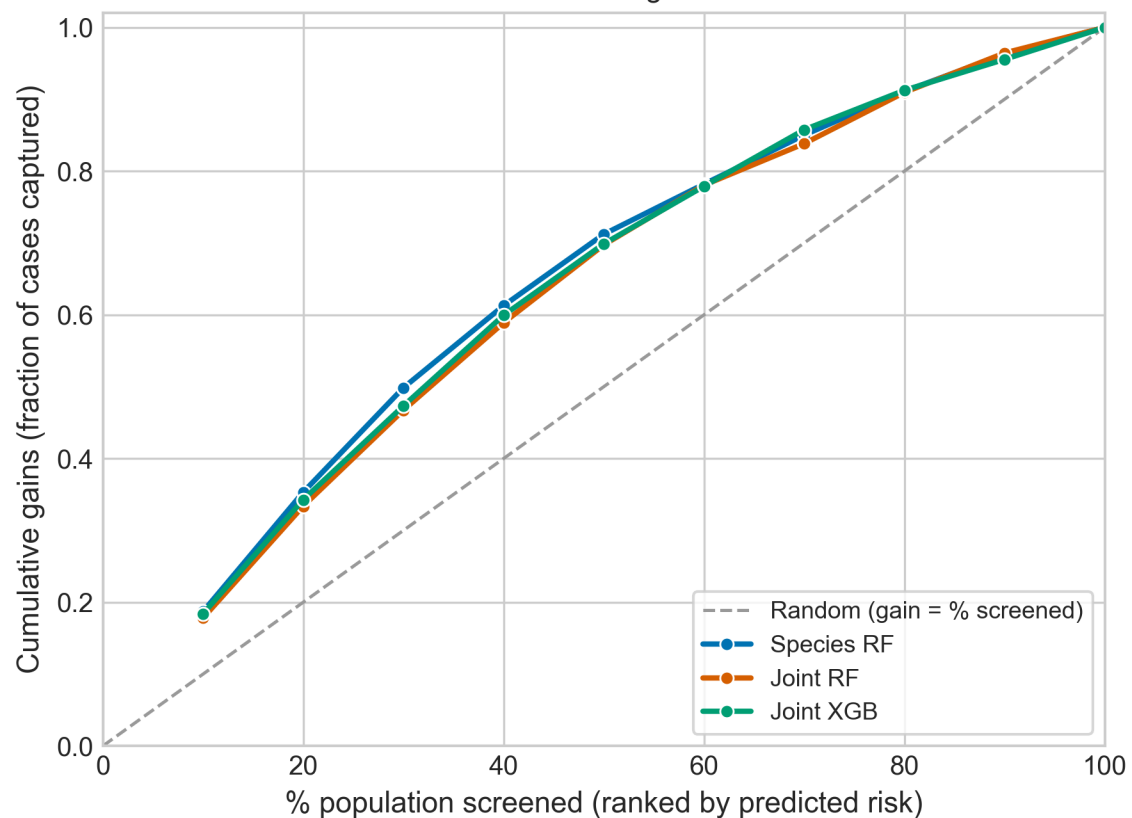


Lift and cumulative gains (pooled LODO predictions, n = 1339)

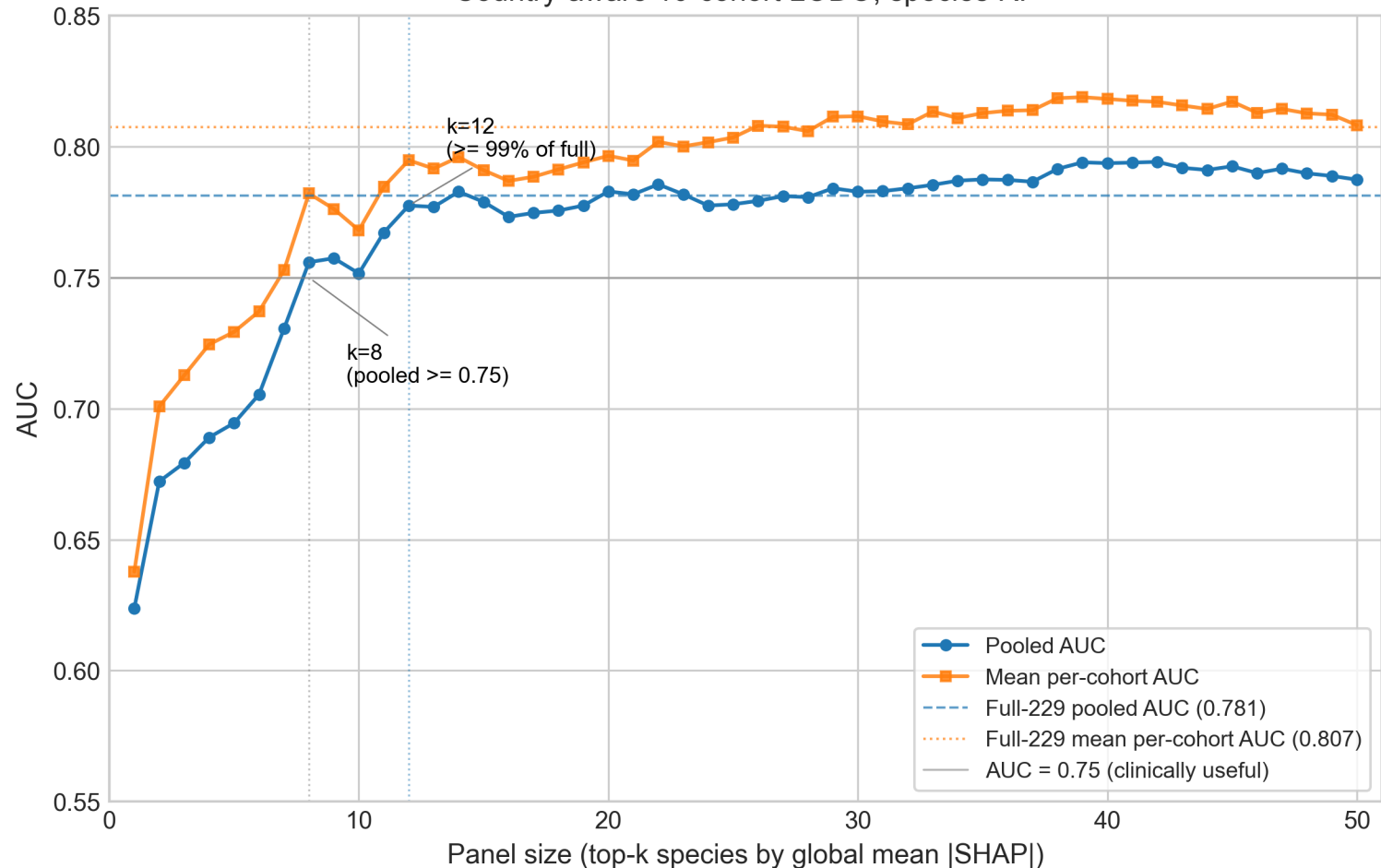
Lift curve



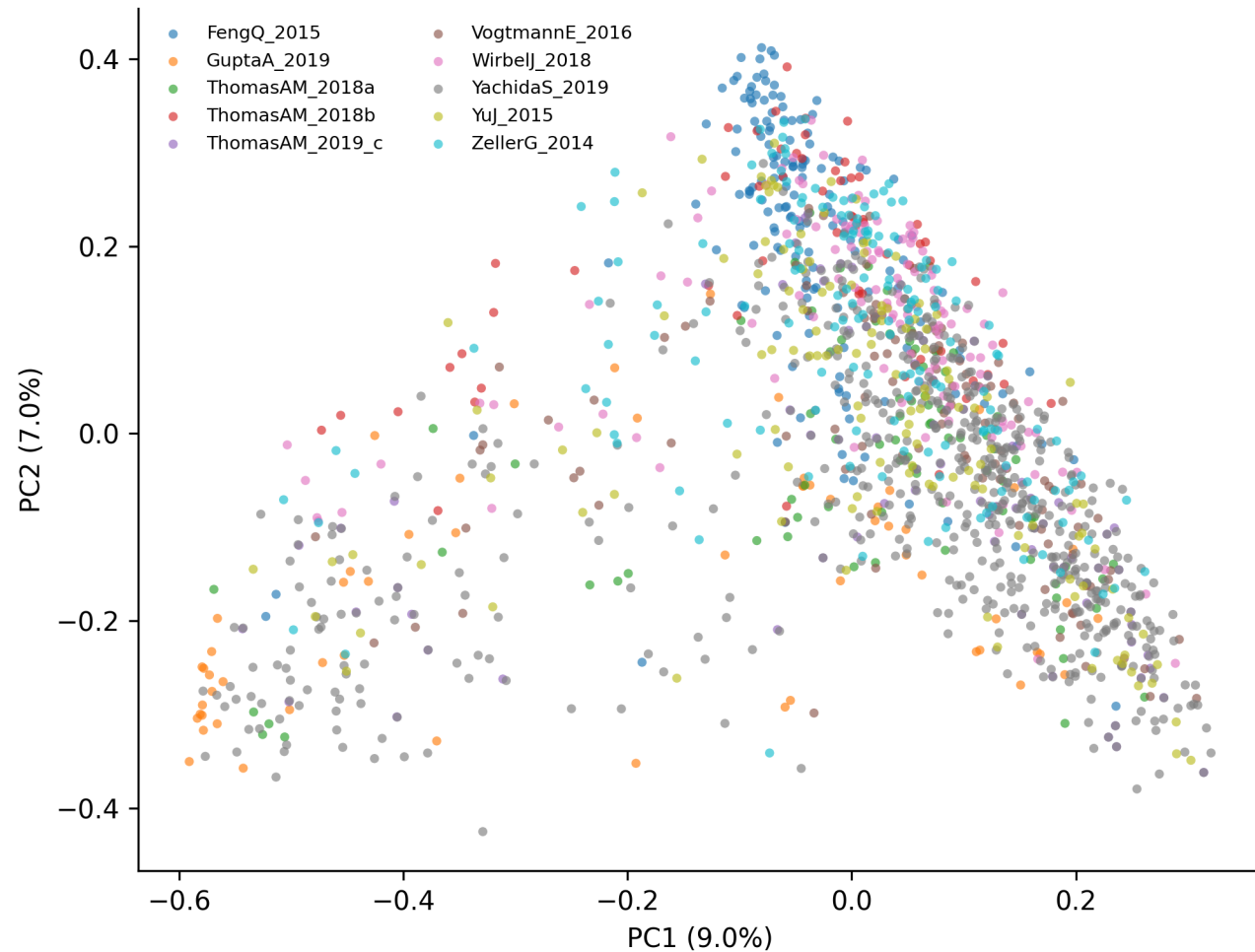
Cumulative gains



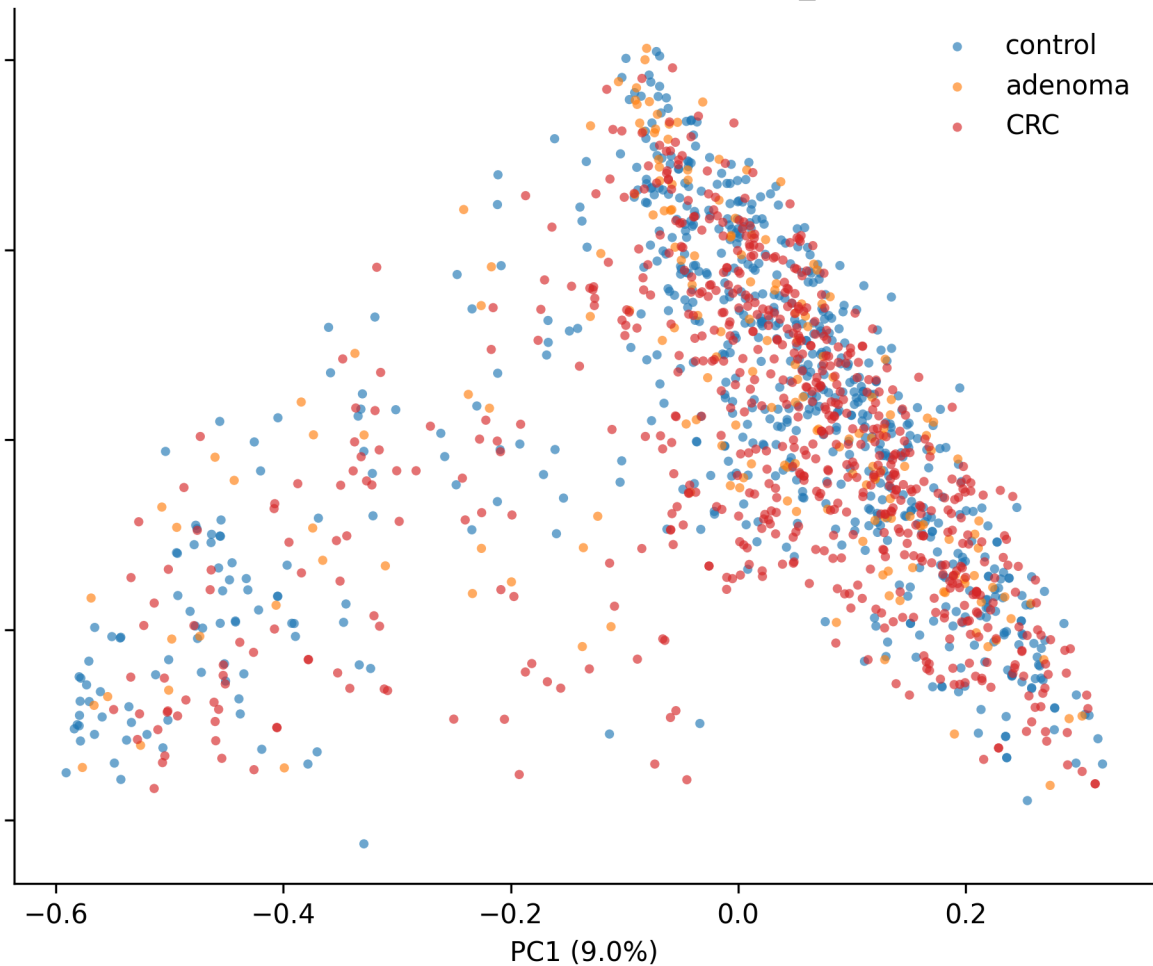
Minimum useful species panel — greedy forward selection
Country-aware 10-cohort LODO, species RF



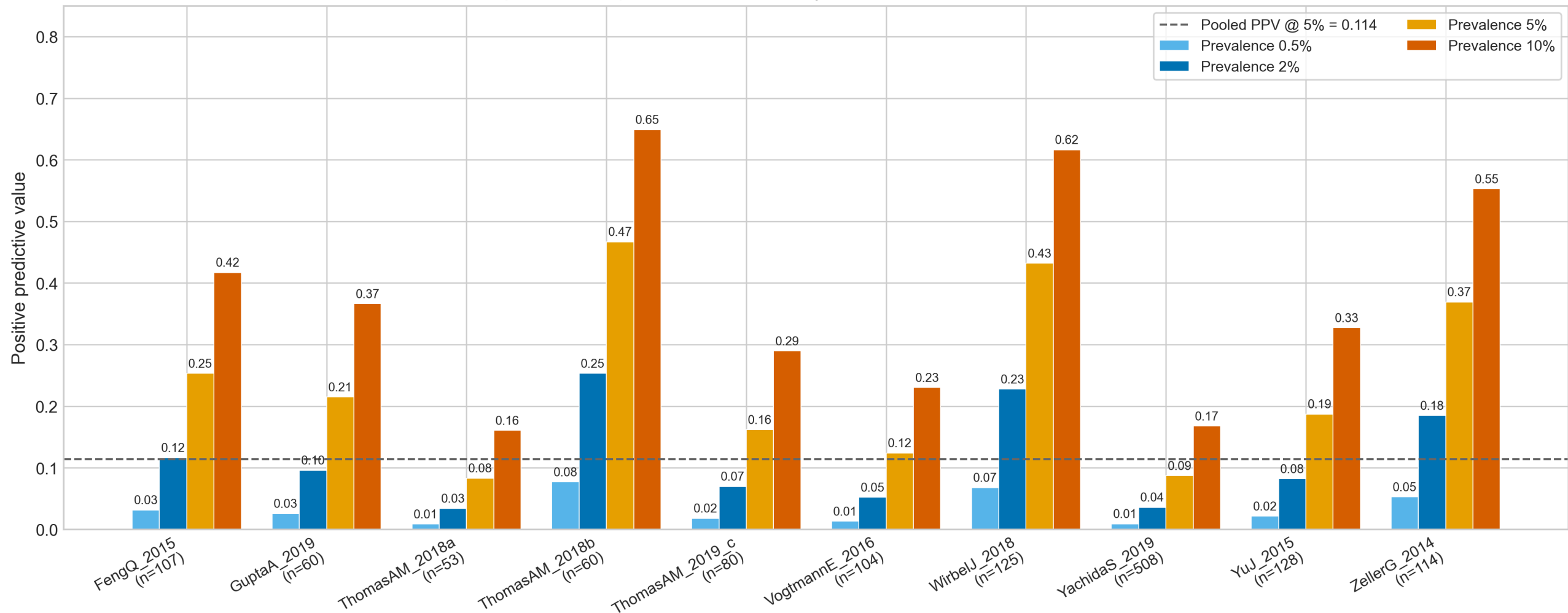
PCoA (Bray-Curtis) coloured by cohort

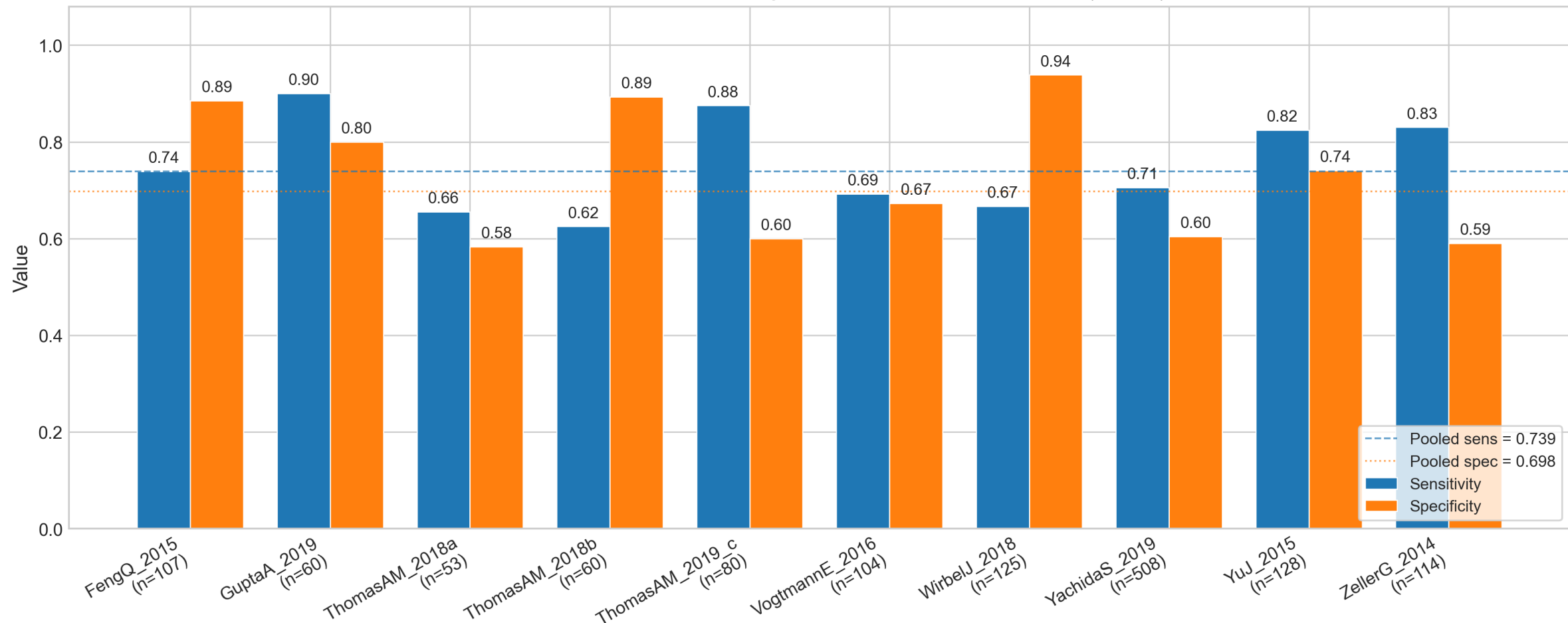


PCoA (Bray-Curtis) coloured by study_condition



Species RF — per-cohort base-rate-adjusted PPV at local Youden-J threshold

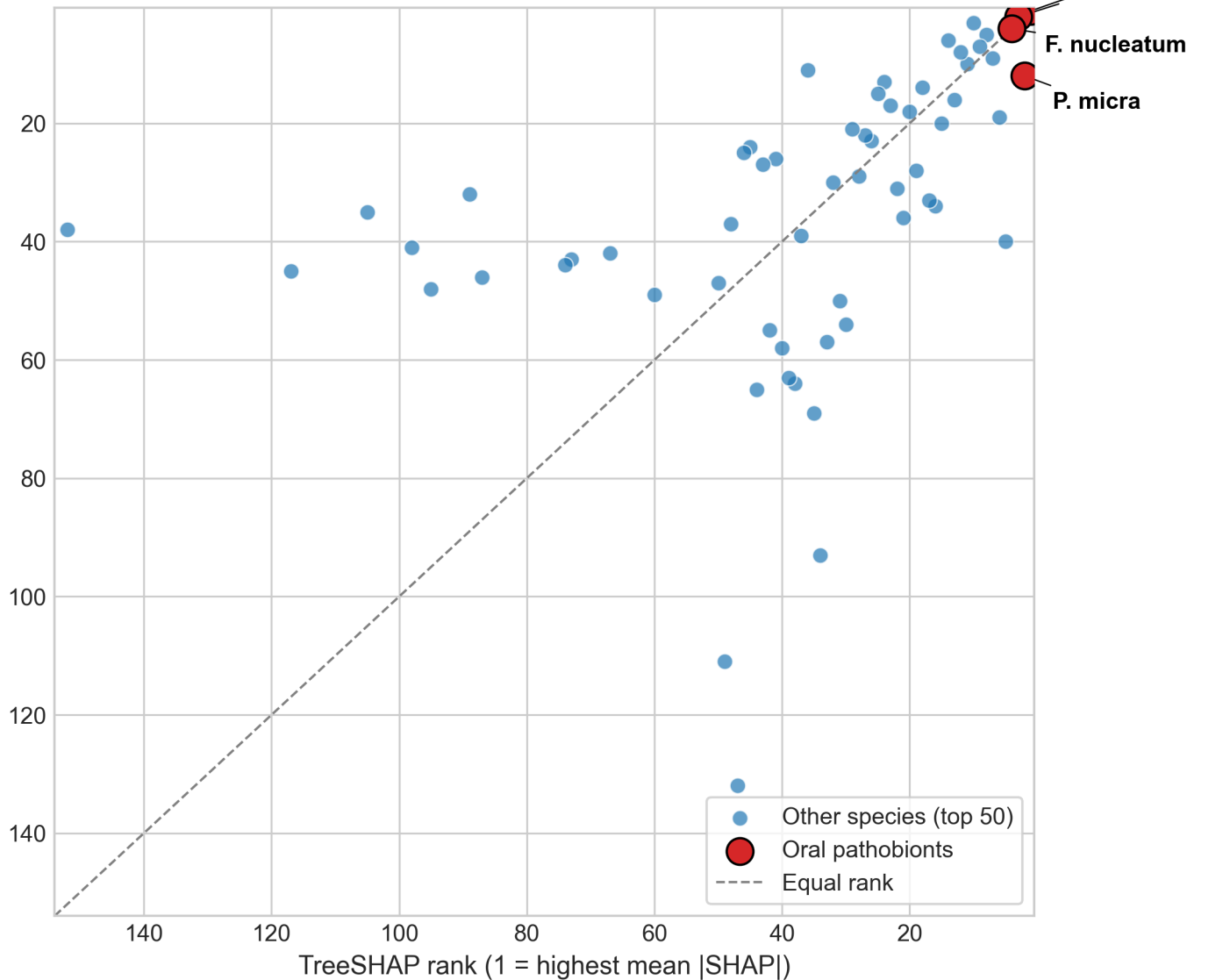


Species RF — per-cohort operating characteristics at pooled Youden-J ($t=0.485$)

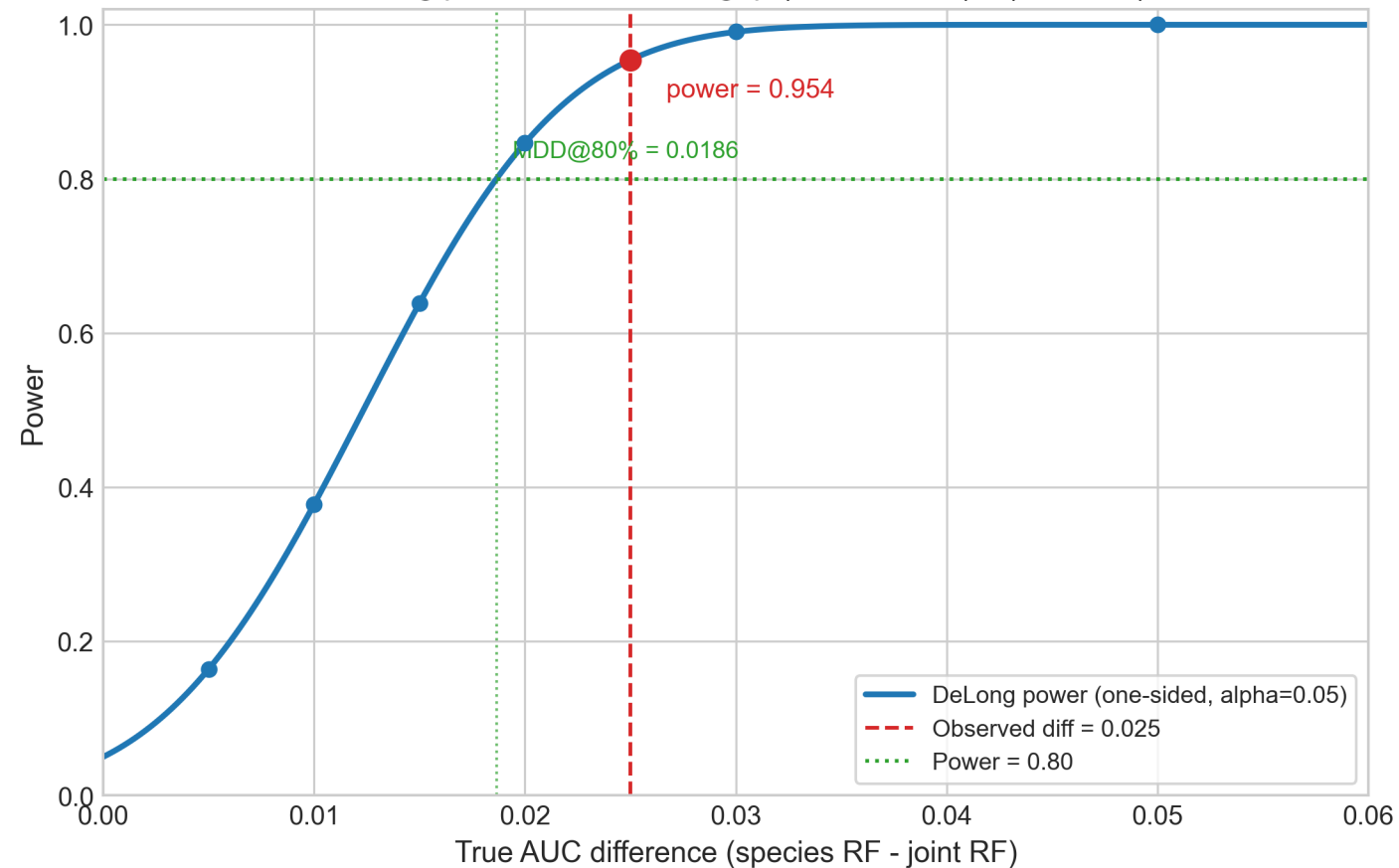
SHAP vs permutation importance — species RF, CRC vs control

Top-20 overlap: 16/20

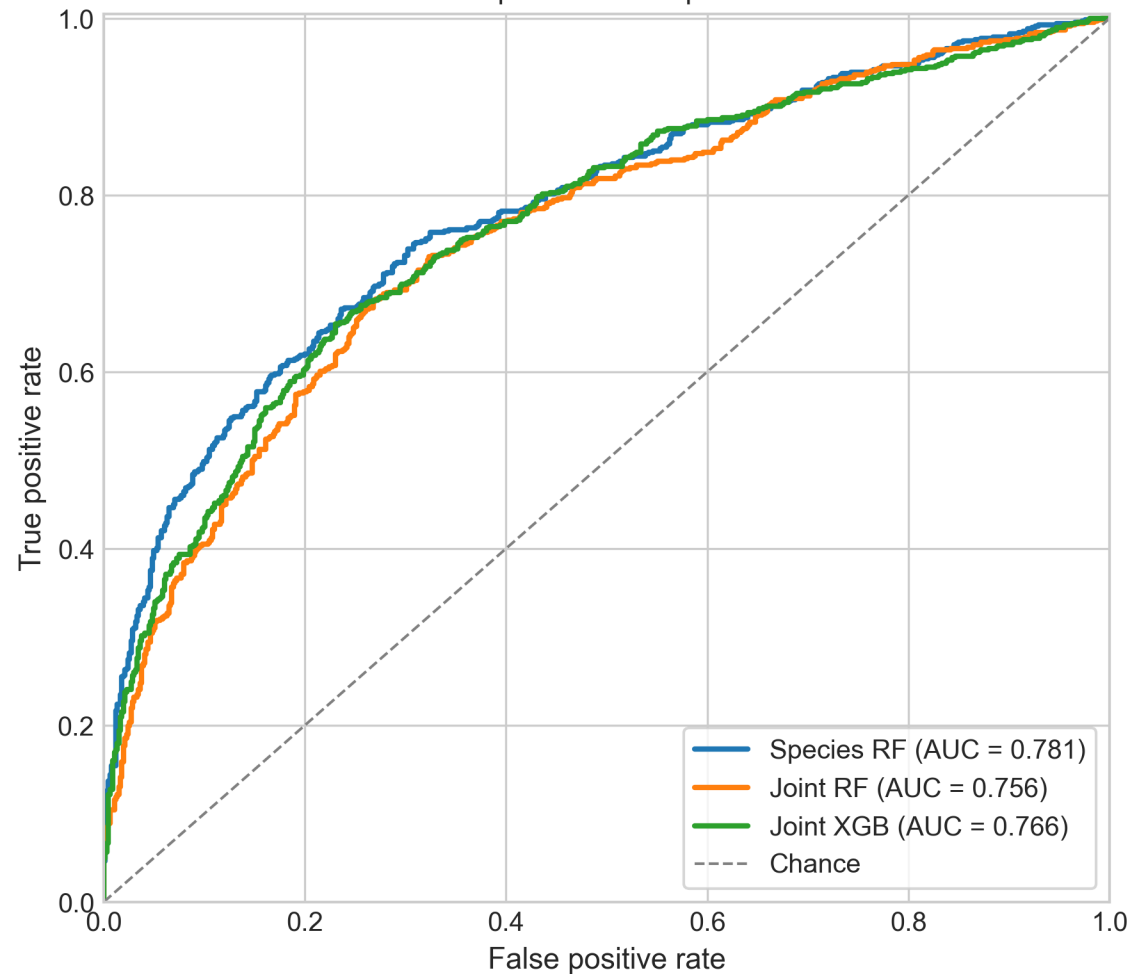
Permutation-importance rank (1 = largest AUC drop)



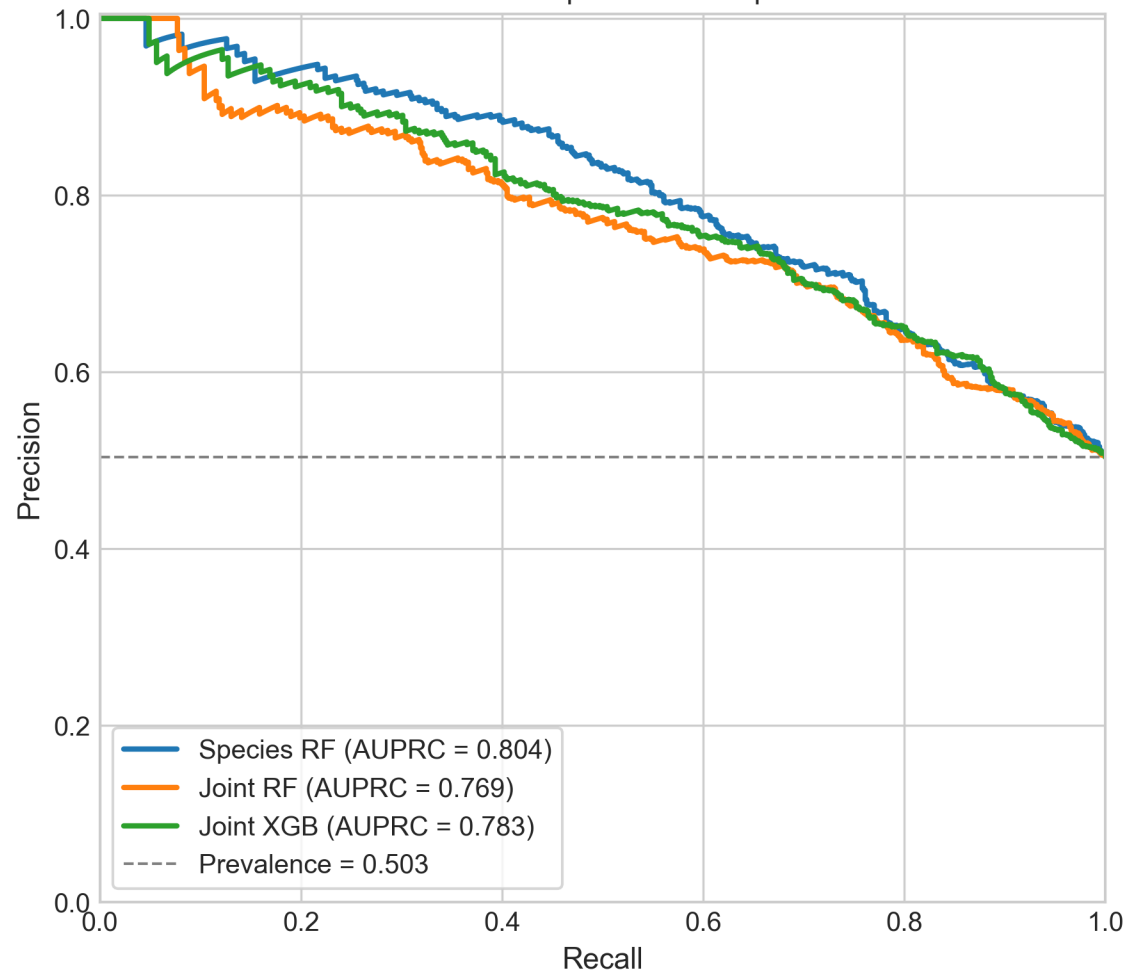
DeLong power vs true AUC gap (n=1339, SE(diff)=0.0075)



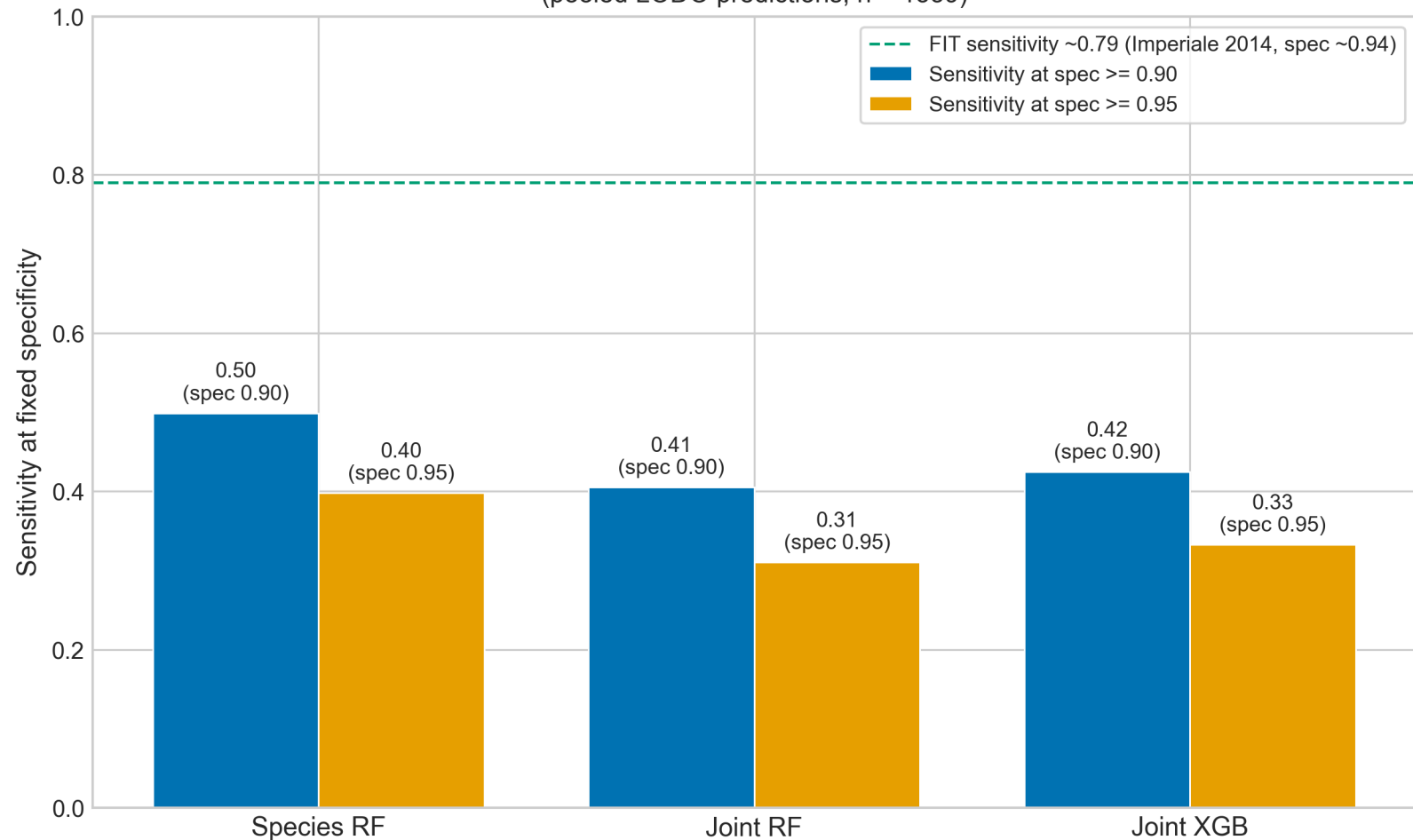
ROC — pooled LODO predictions



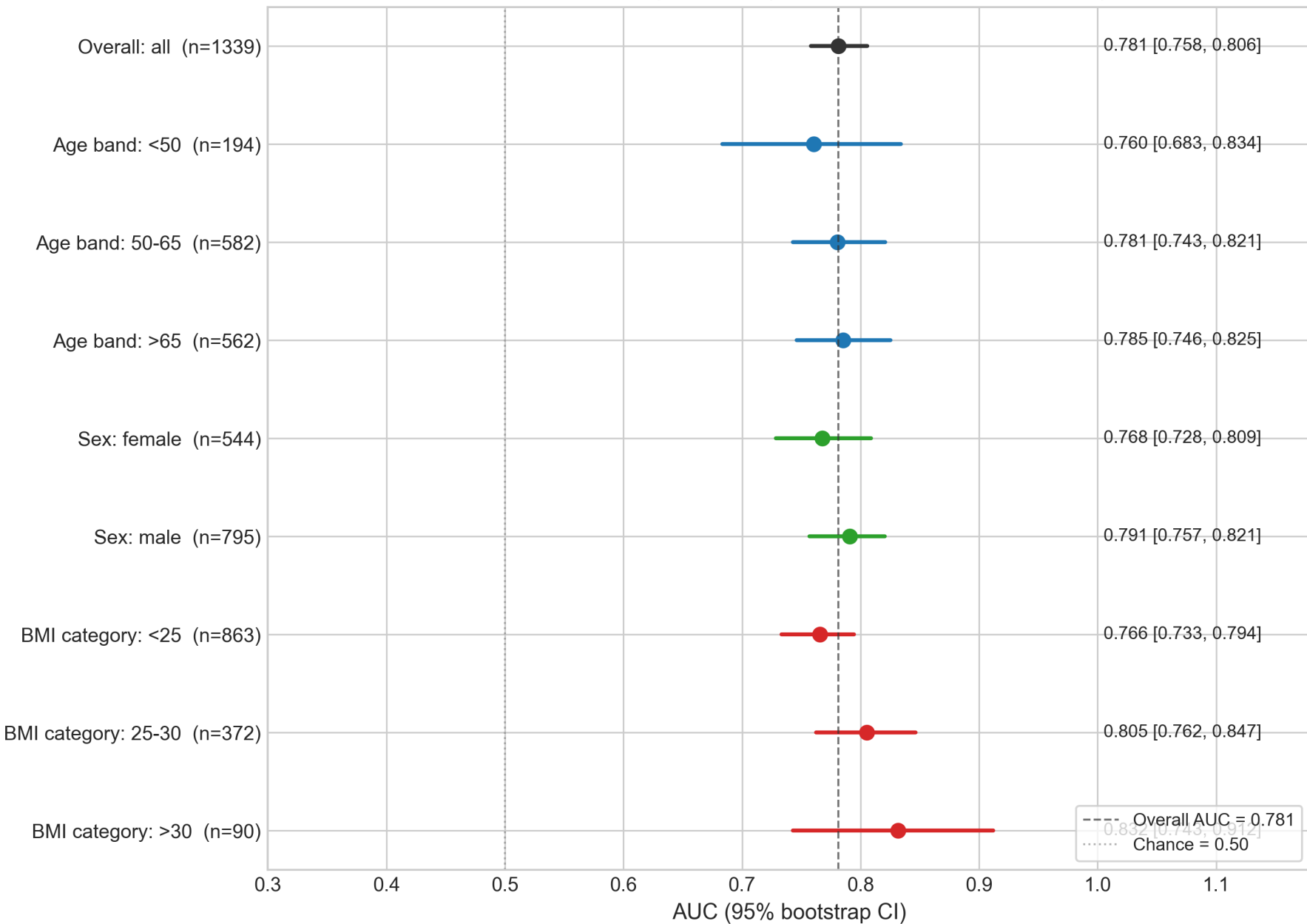
Precision-Recall — pooled LODO predictions



Sensitivity at clinically-meaningful specificity floors
(pooled LODO predictions, n = 1339)



Species RF — subgroup AUC (pooled LODO, 1000-iter bootstrap)



A heatmap visualization showing the relative abundance of various bacterial taxa over time. The y-axis lists the taxa names, rotated diagonally for readability: Prevotella copri, Bacteroides vulgatus, Faecalibacterium prausnitzii, Bacteroides uniformis, Ruminococcus bromii, Escherichia coli, Alistipes putredinis, Eubacterium rectale, Bacteroides dorei, Bifidobacterium adolescentis, Bacteroides stercoris, Collinsella aerofaciens, Eubacterium sp CAG 180, Roseburia faecis, Bifidobacterium longum, Parabacteroides distasonis, Eubacterium eligens, Akkermansia muciniphila, Anaerostipes hadrus, and Bacteroides plebeius. The x-axis represents time points: 2015, 2019, 2018a, 2018b, 2019_c, 2016, 2018, 2019, 2015, and 2014. Each cell's color indicates its relative abundance, ranging from light yellow/green (low) through green to dark purple/black (high). Notable features include a bright yellow-green spot at the intersection of P. copri and 2019; several black spots indicating very low abundance, such as those for R. faecis at 2018a and 2019; and generally higher abundances (green/yellow) for many taxa in the earlier years like 2015 and 2014 compared to later ones like 2018a and 2018b.

